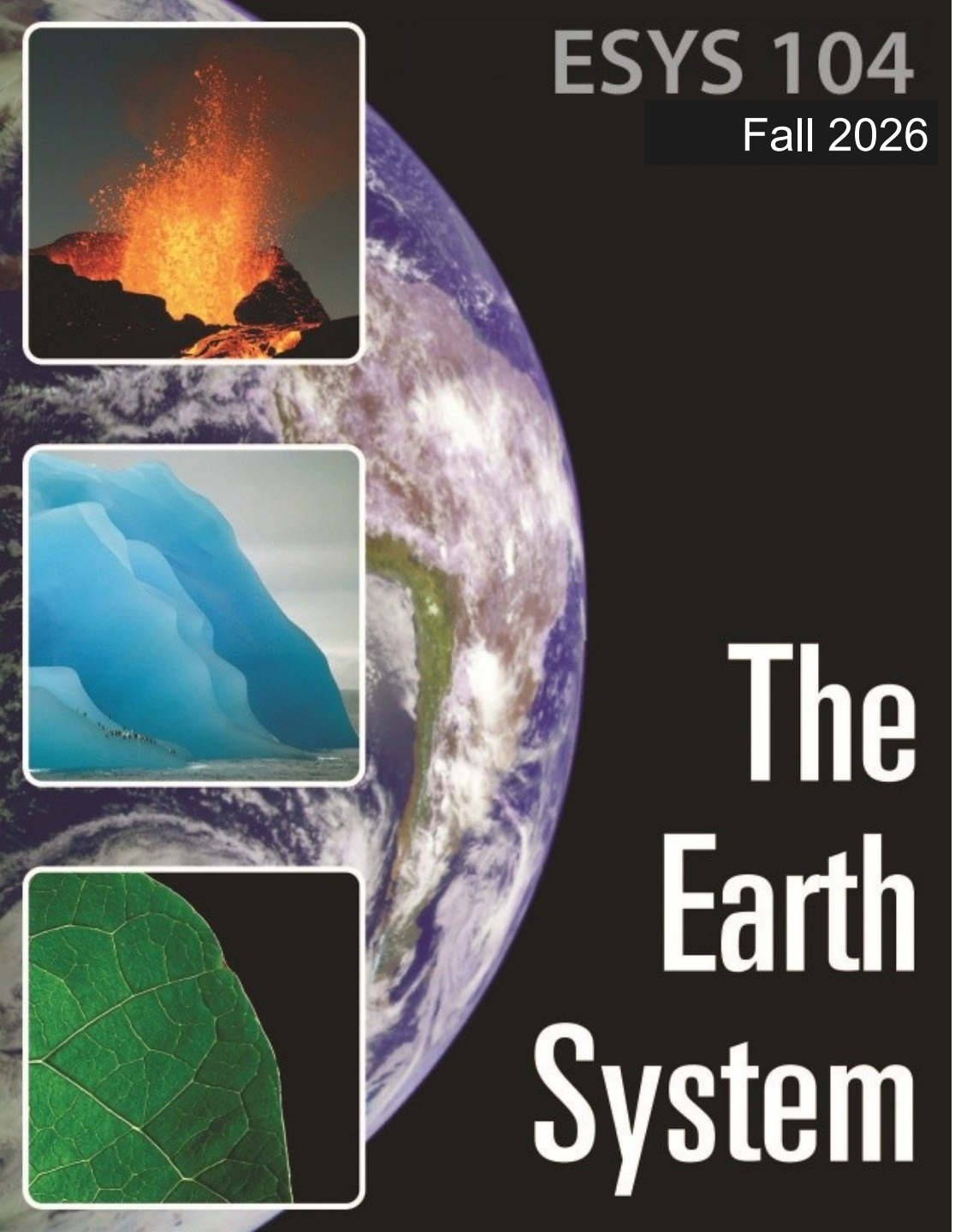


#8 Deep Ocean circulation and the polar areas

Hydrological cycle readings posted

Assignment #1 due tonight

Readings: Lectures 9, 10, 11
(in Readings)

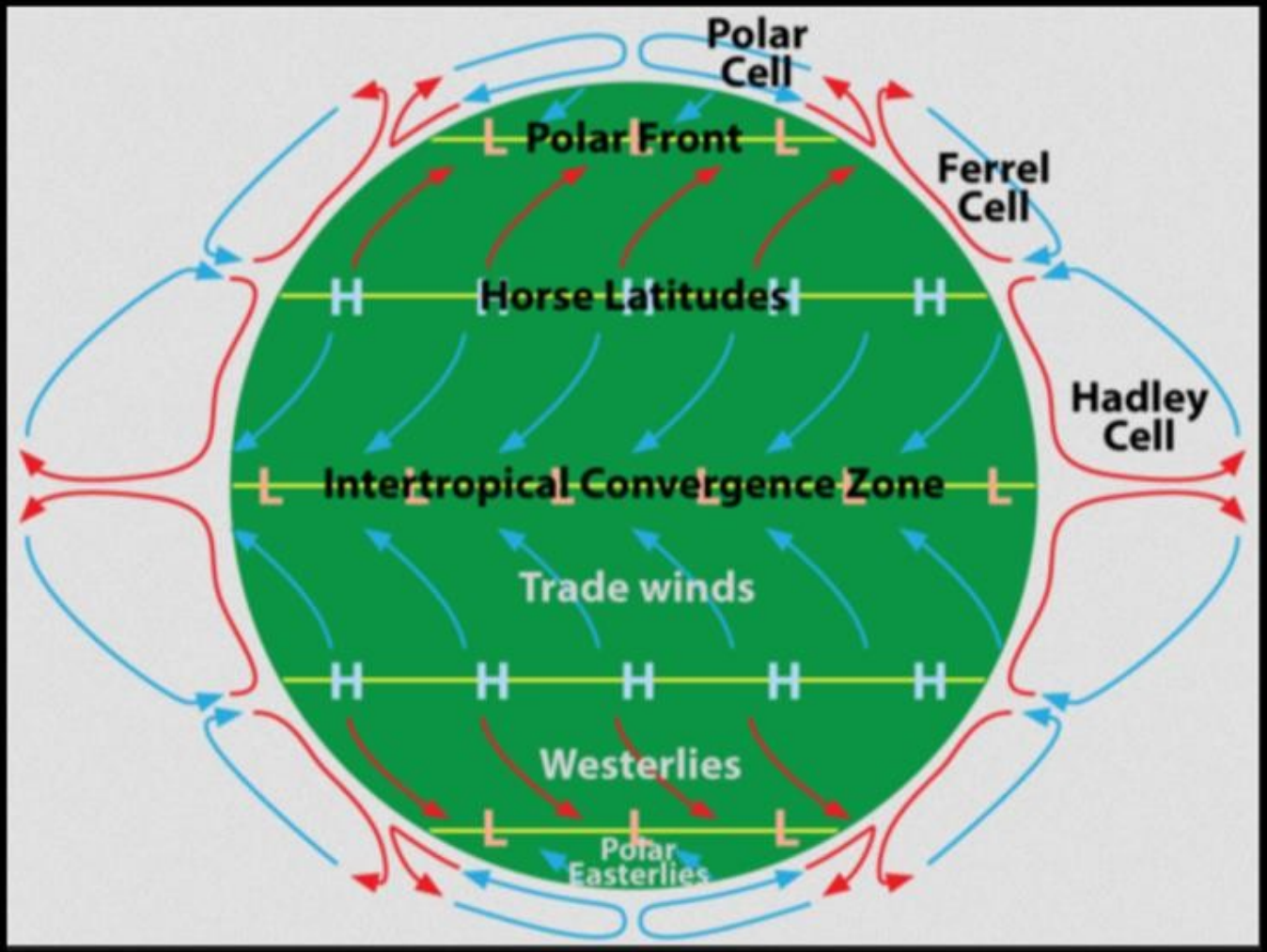
Strahler and Strahler Chapter 9 (pp 161-170) and Chapter 16 (pp 298-312)

And Ecosystems:
"Ecosystem Services" from the Ecological Society of America

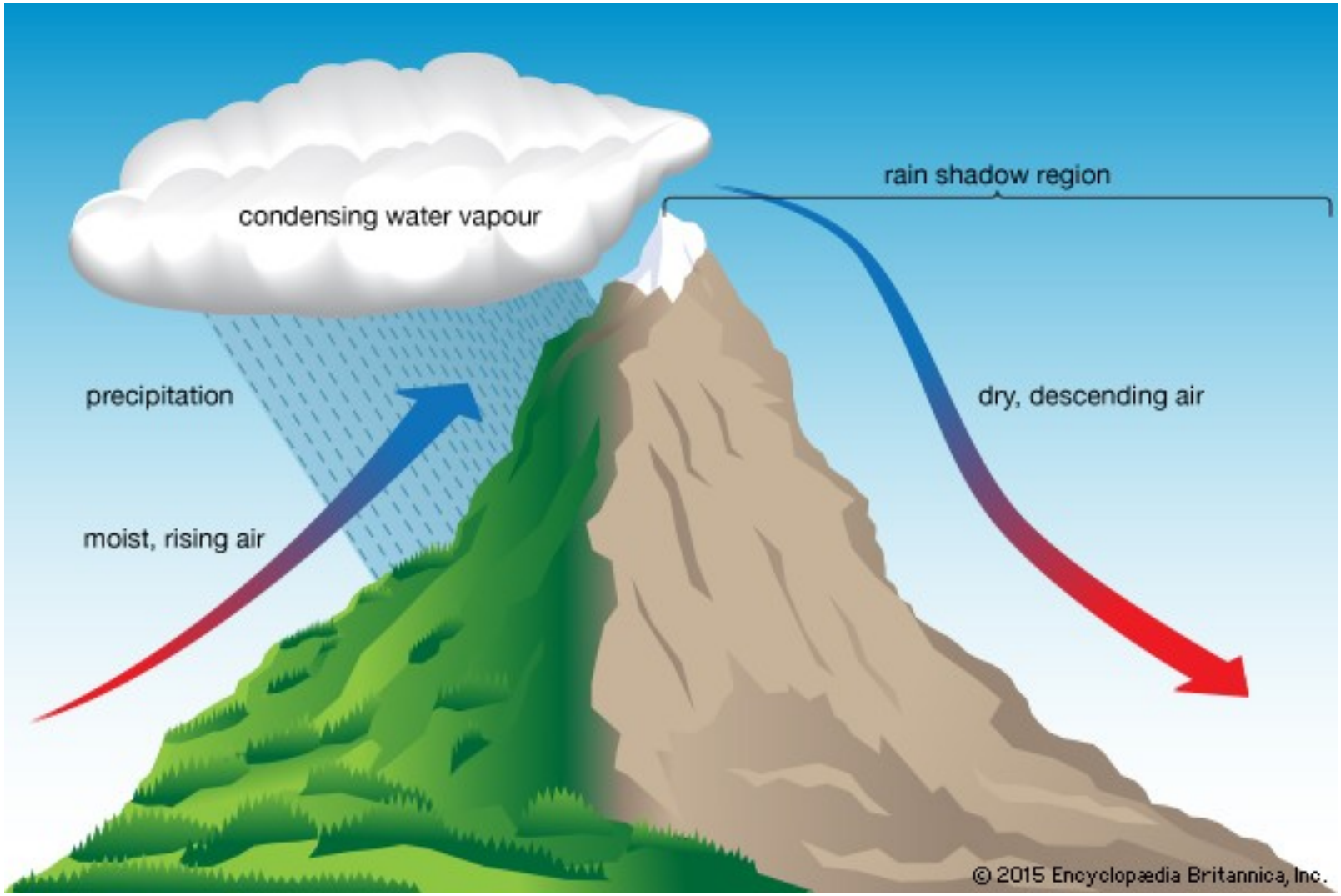
For Lecture 11: add:
REGIME SHIFTS, RESILIENCE, AND BIODIVERSITY IN ECOSYSTEM MANAGEMENT
Carl Folke, 2004

Foley, 2003: Regime Shift/Sahel

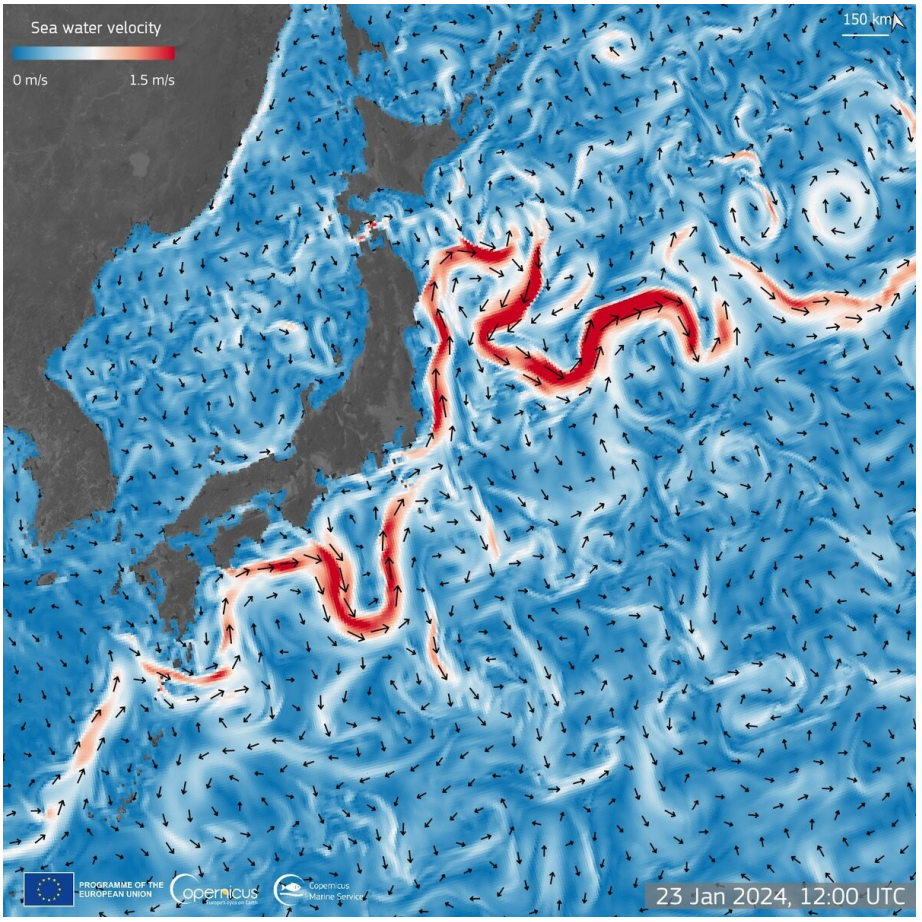
Atmospheric circulation controls mean precipitation levels over various parts of globe

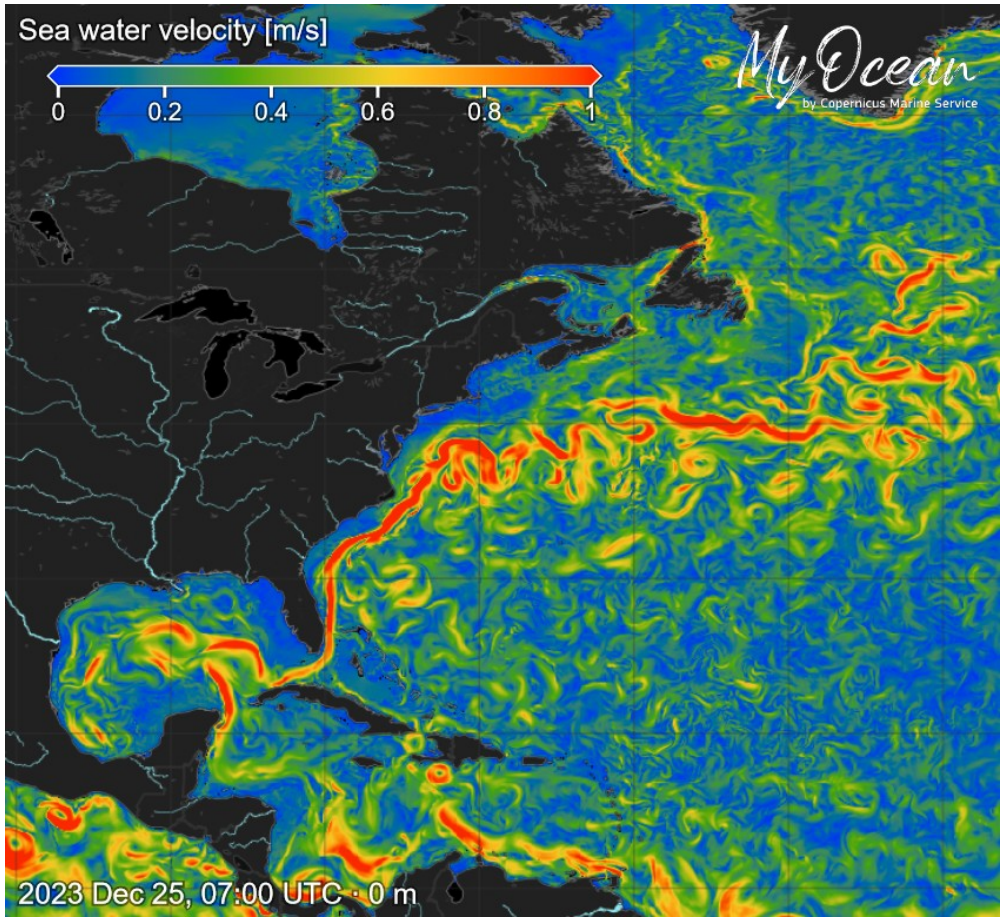


Effect of some mountain belts on rainfall patterns: “orographic rainfall”



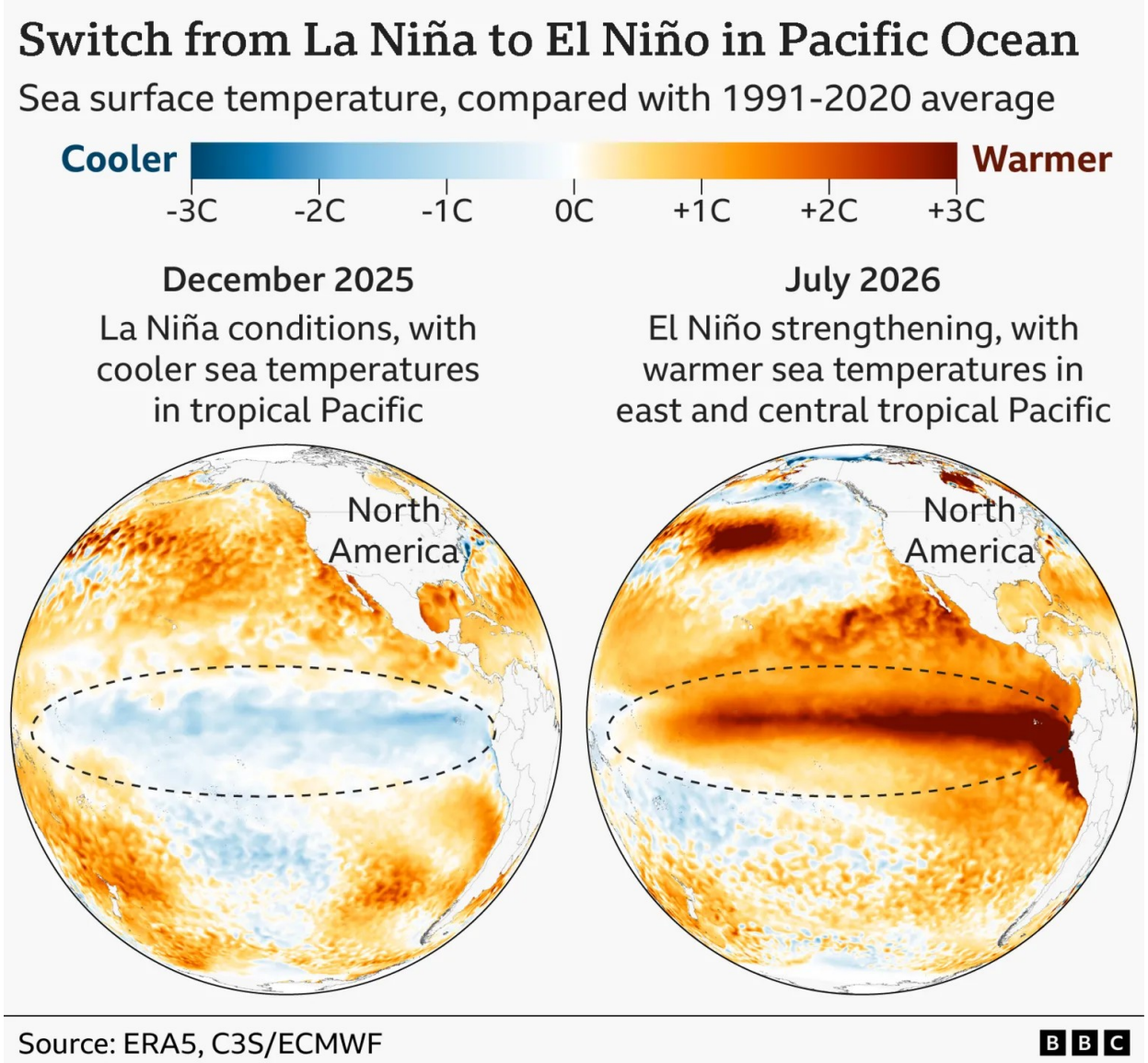
Surface Ocean Circulation





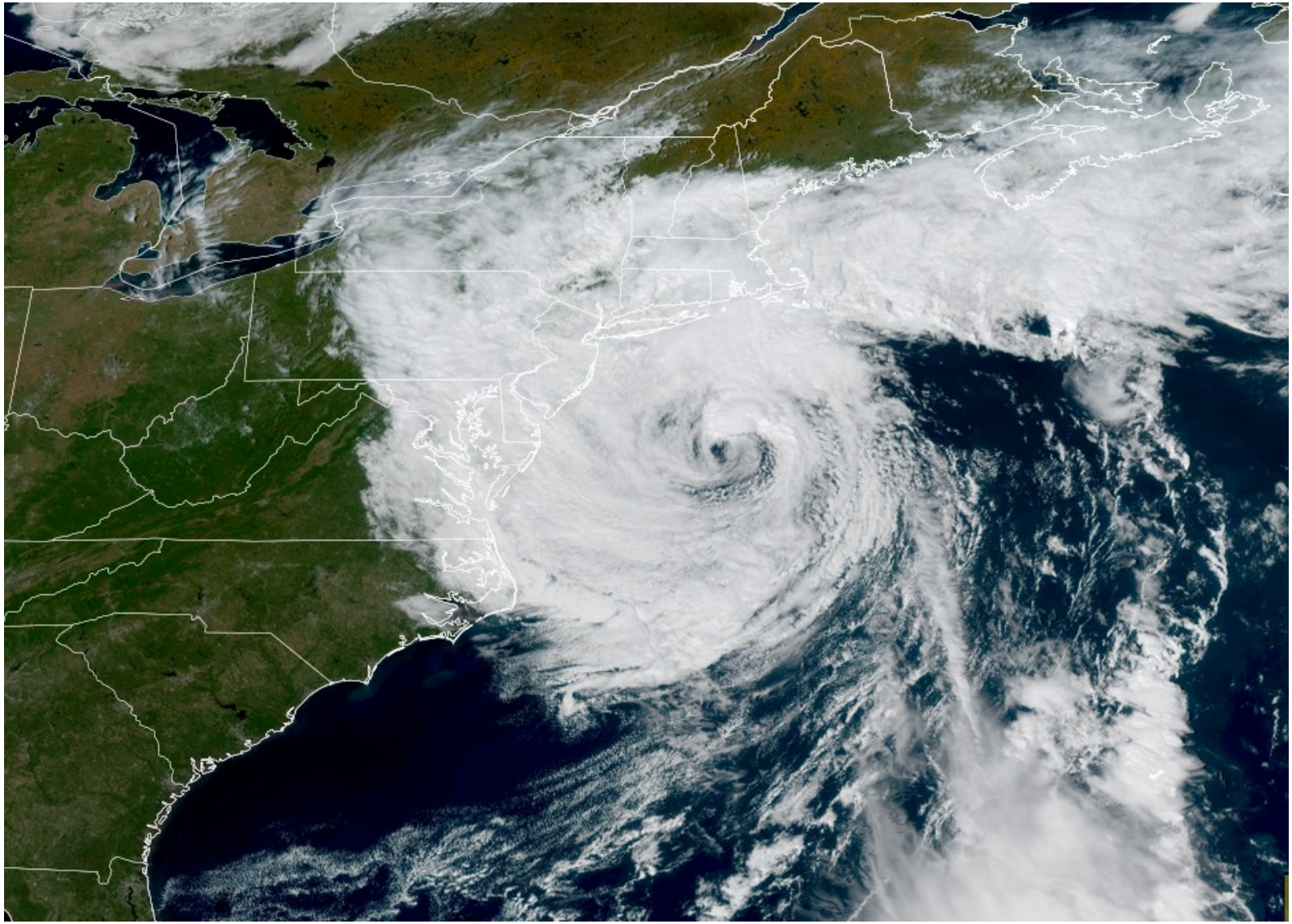
Kuroshio and Gulf Stream Currents

El Niño



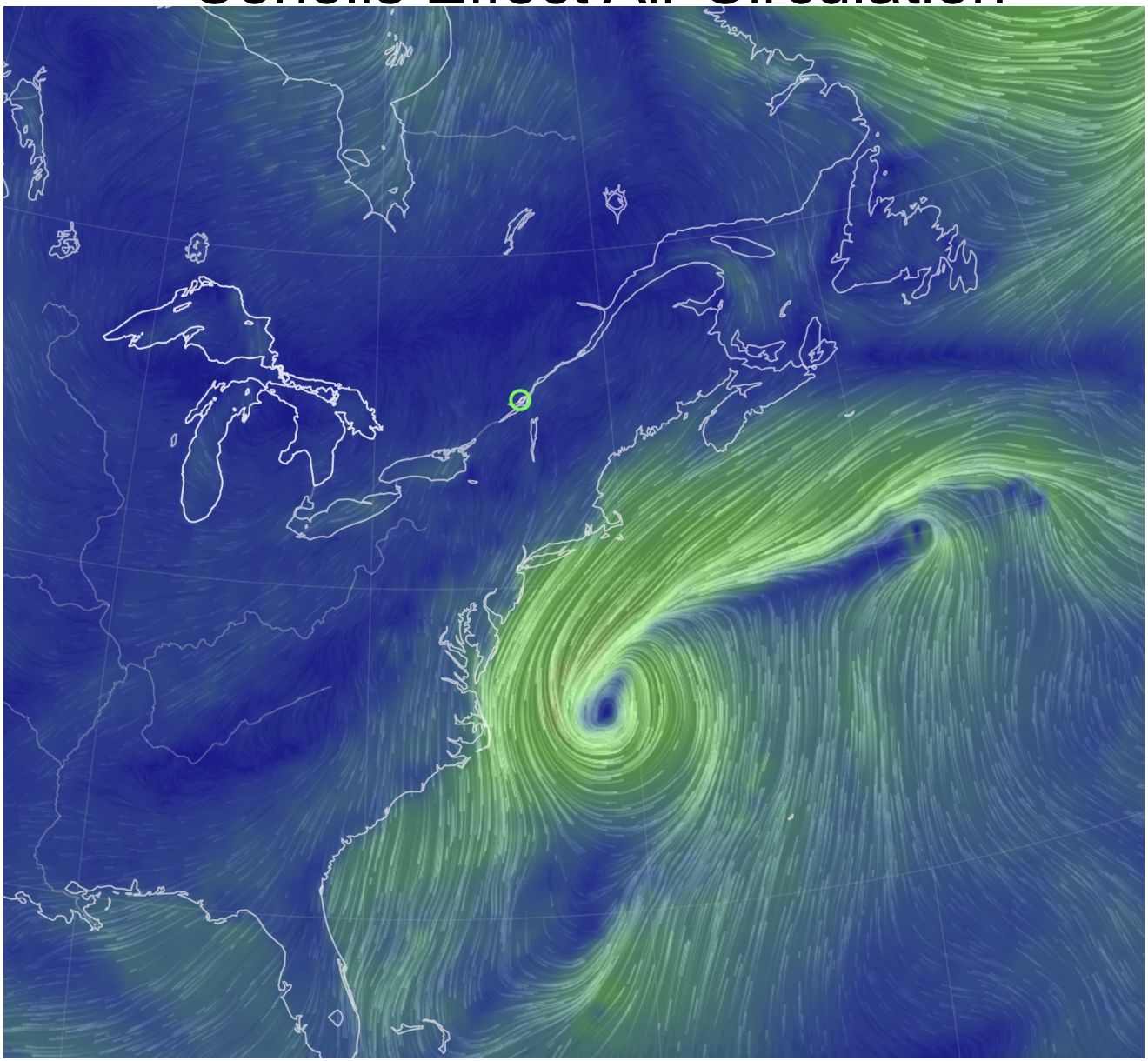
Perturbations from the normal pattern

Coriolis Effect Air Circulation



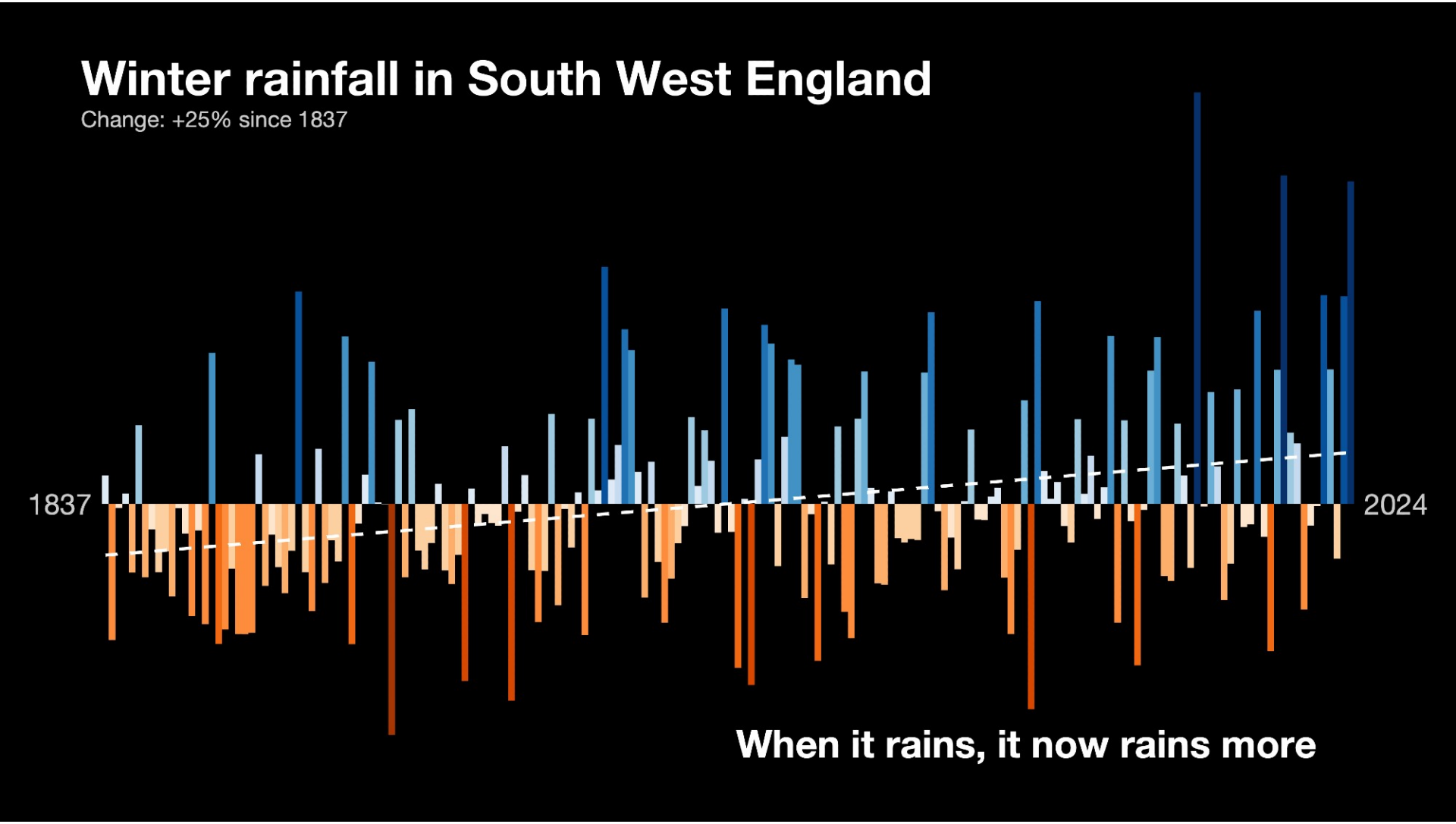
Mid-Latitude Storm Systems

Coriolis Effect Air Circulation

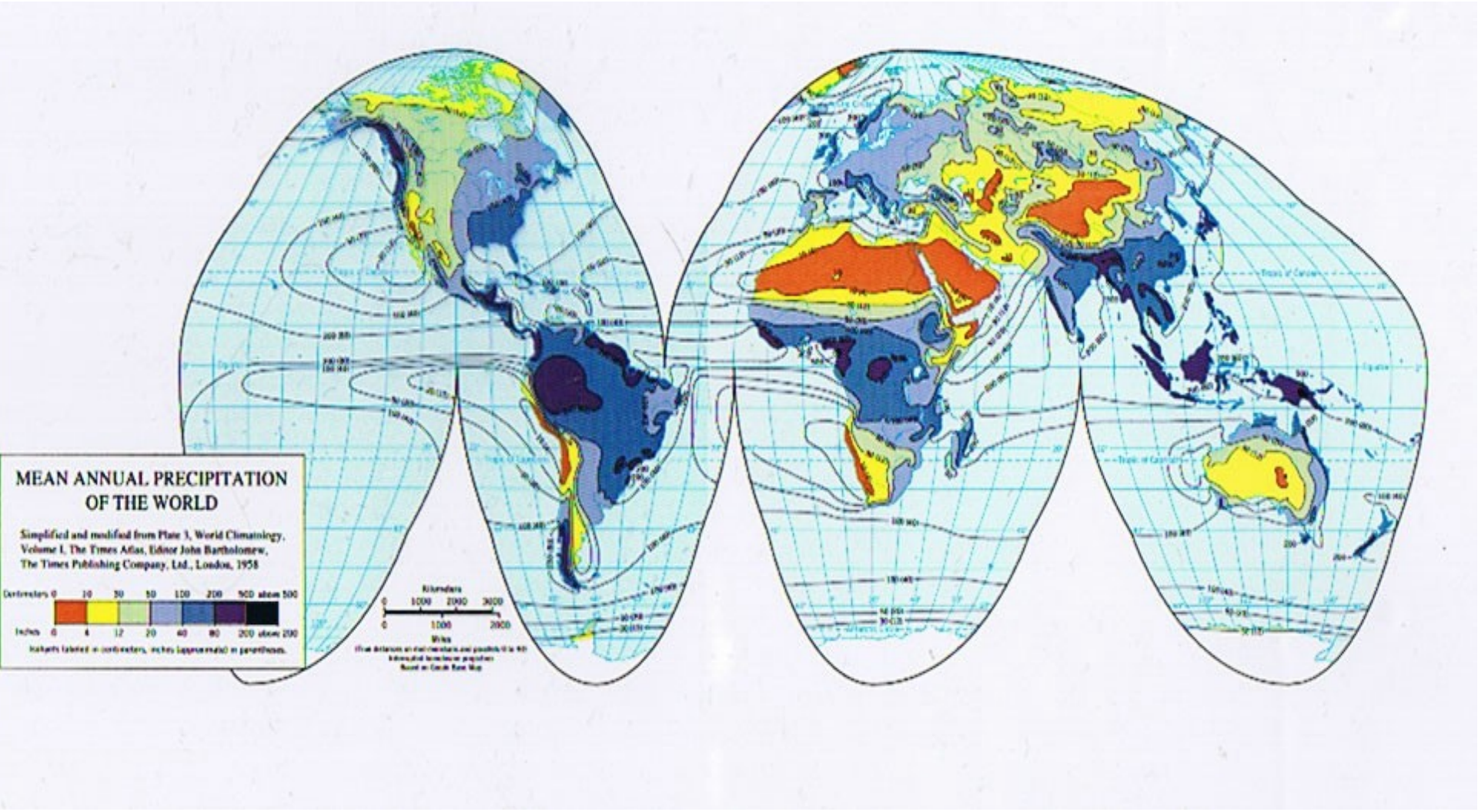


Mid-Latitude Storm Systems

England Winter Rainfall

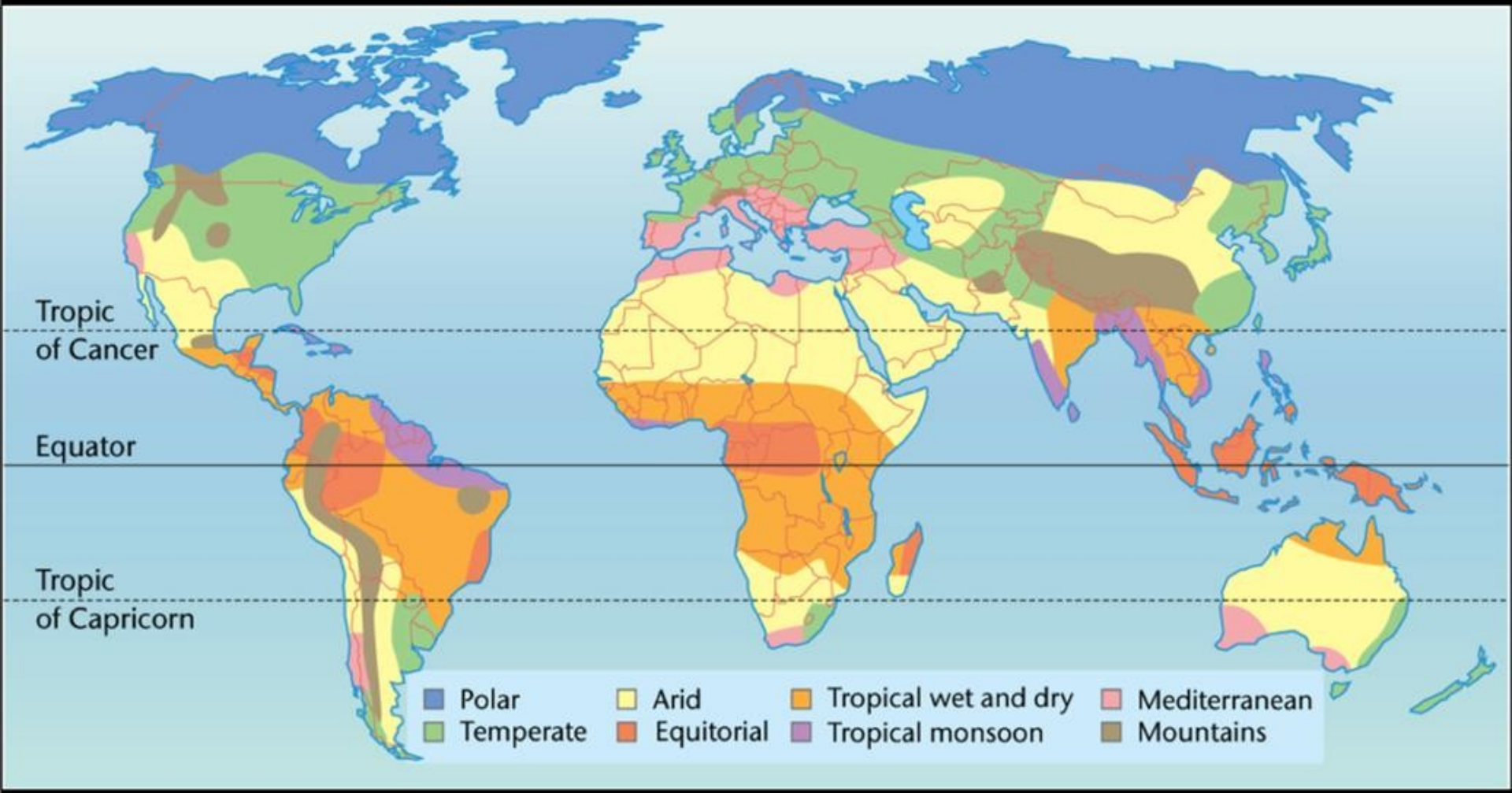


Worldwide average annual precipitation

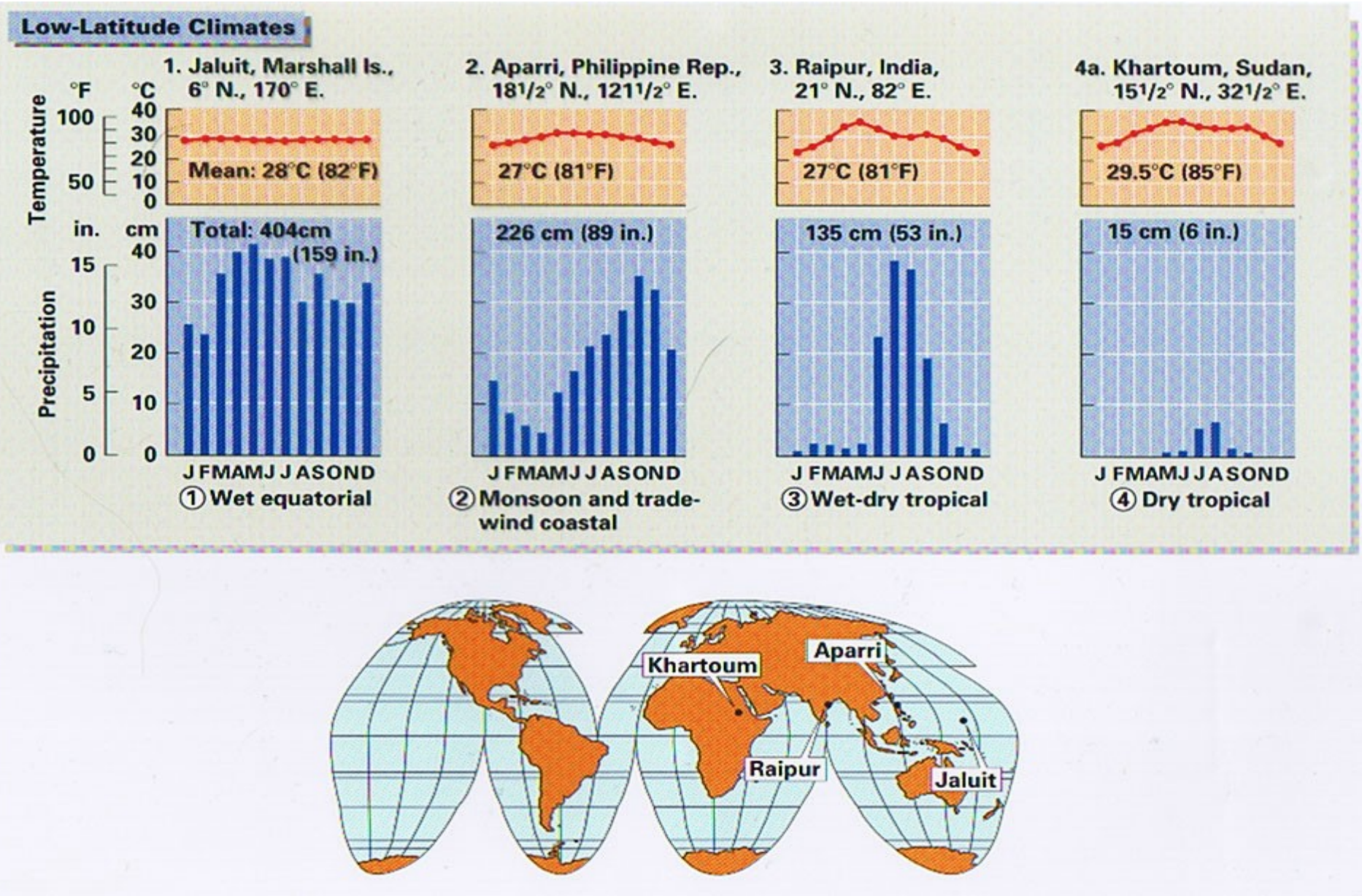


Source: Strahler and Strahler, 2005 Physical Geography, 3rd Edition

Climate Zones

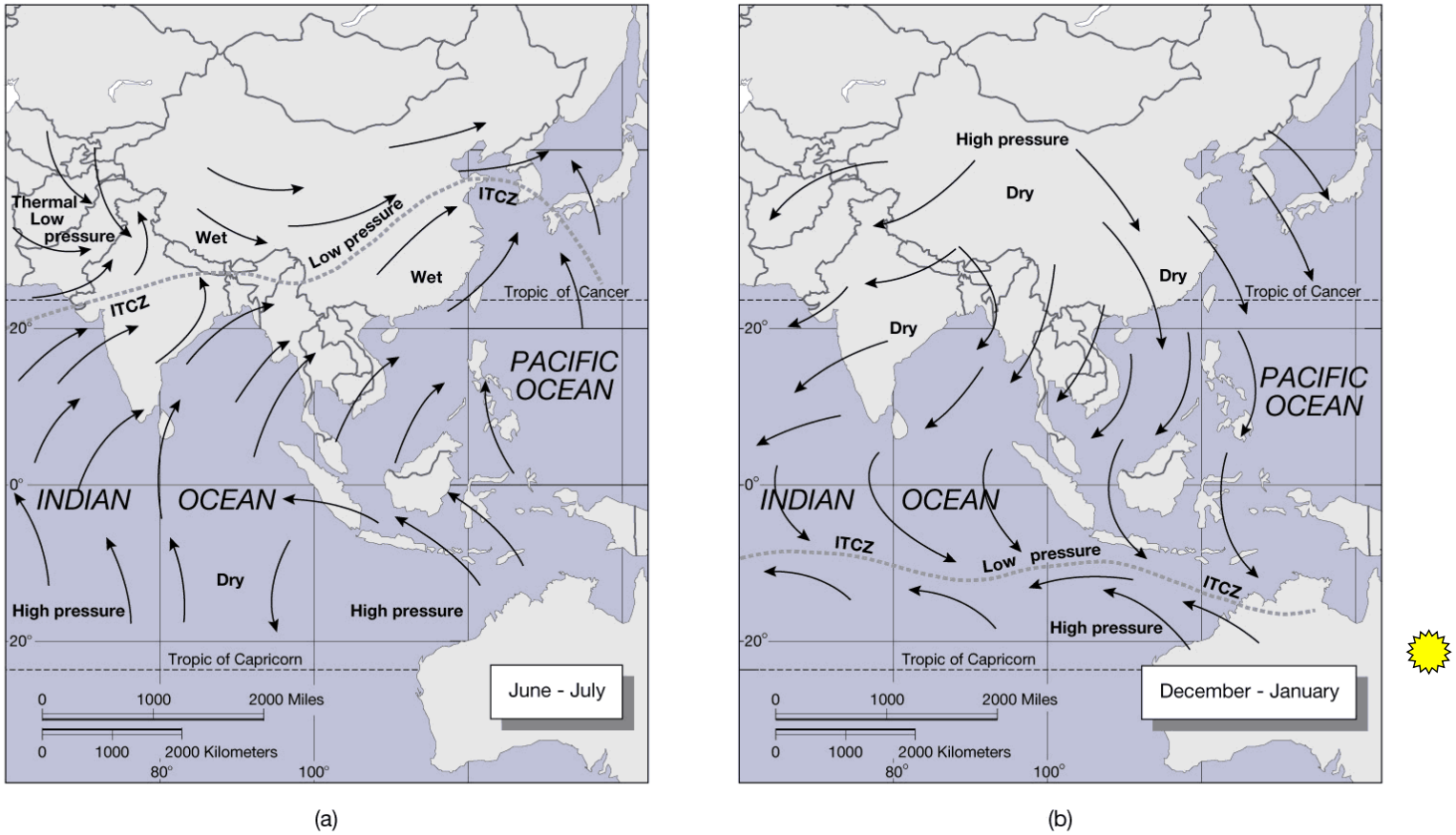


Location within the Global Cells: all four between 6 and 21°N

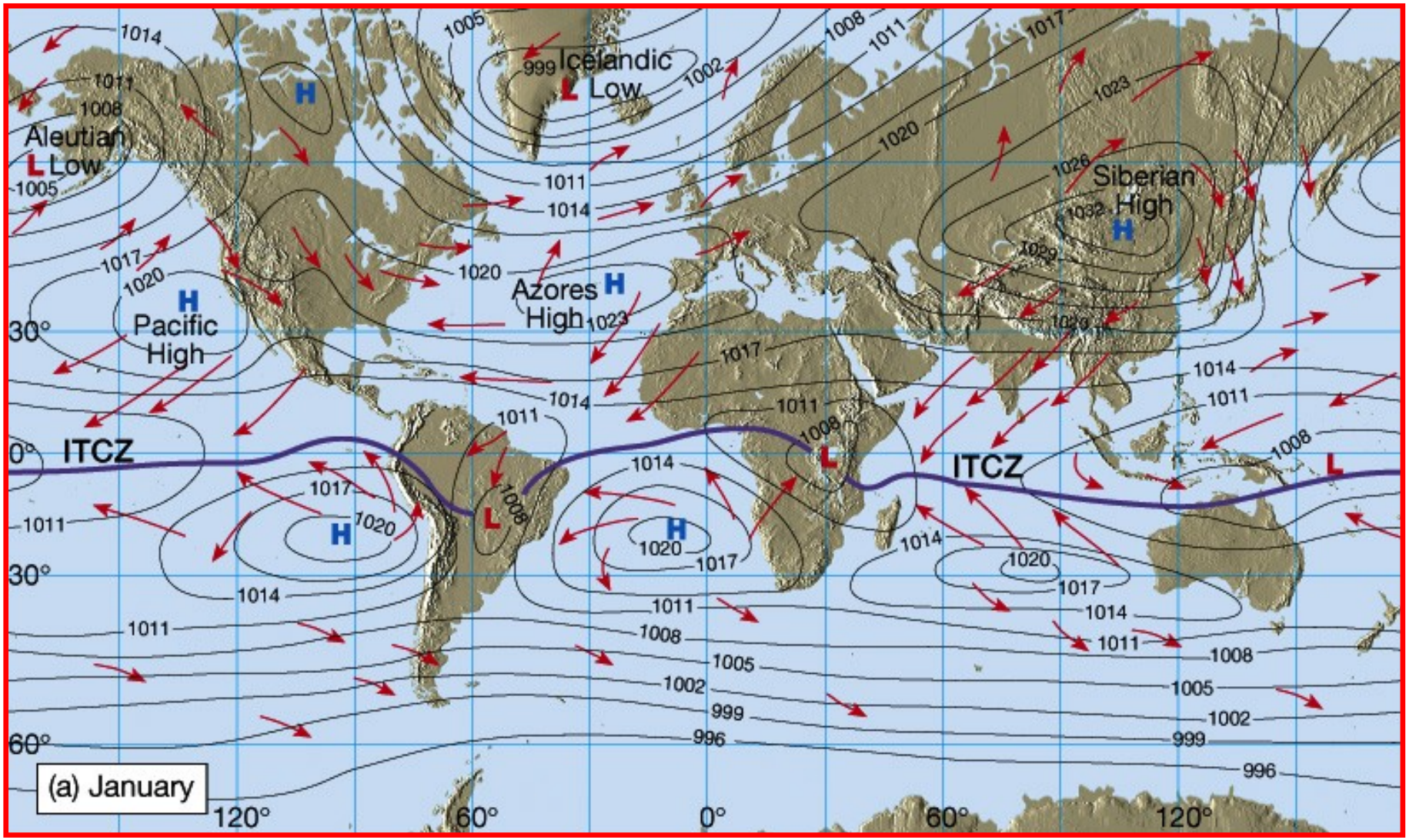


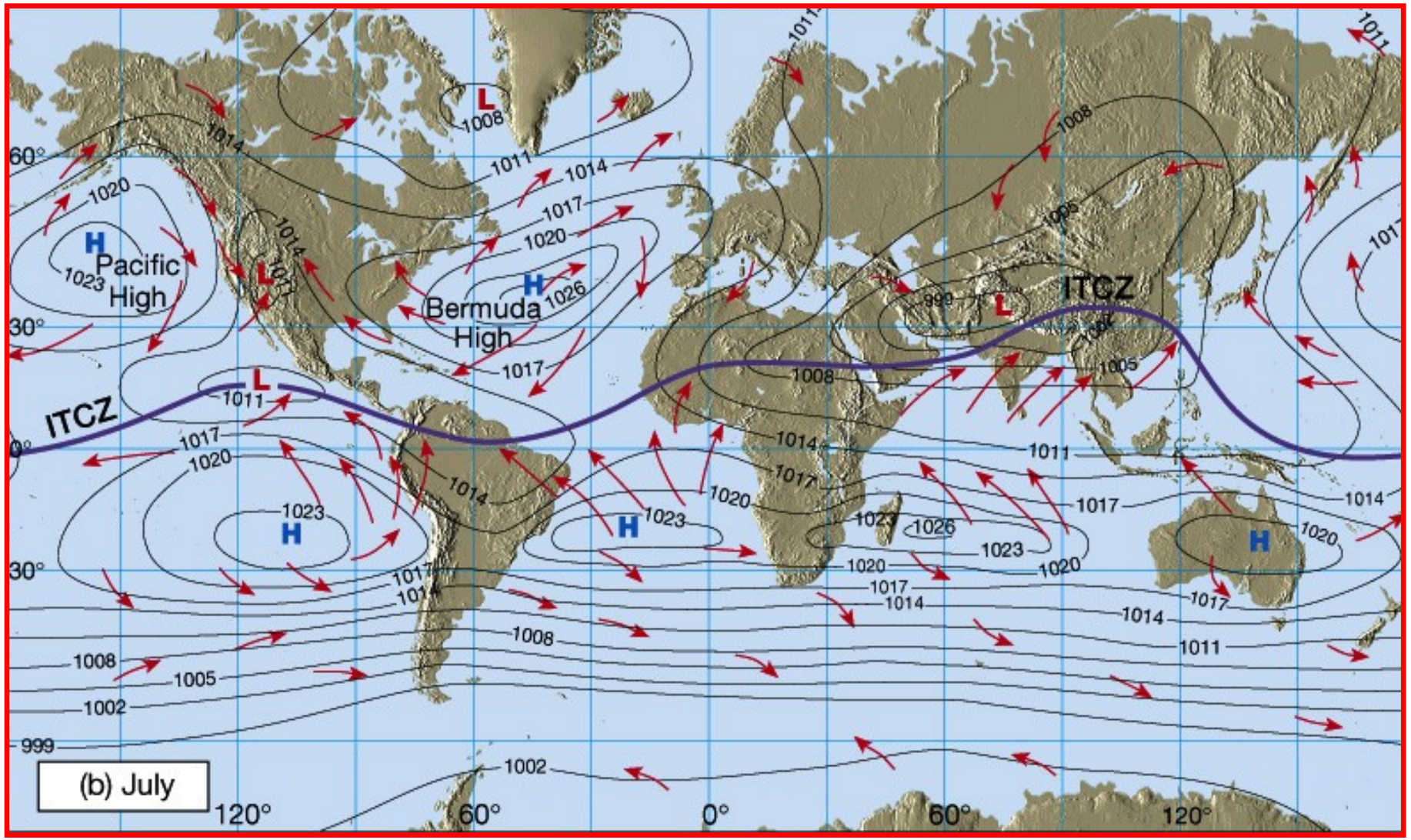
Source: Strahler and Strahler, 2005 Physical Geography, 3rd Edition

Air Temperature-Driven Circulations



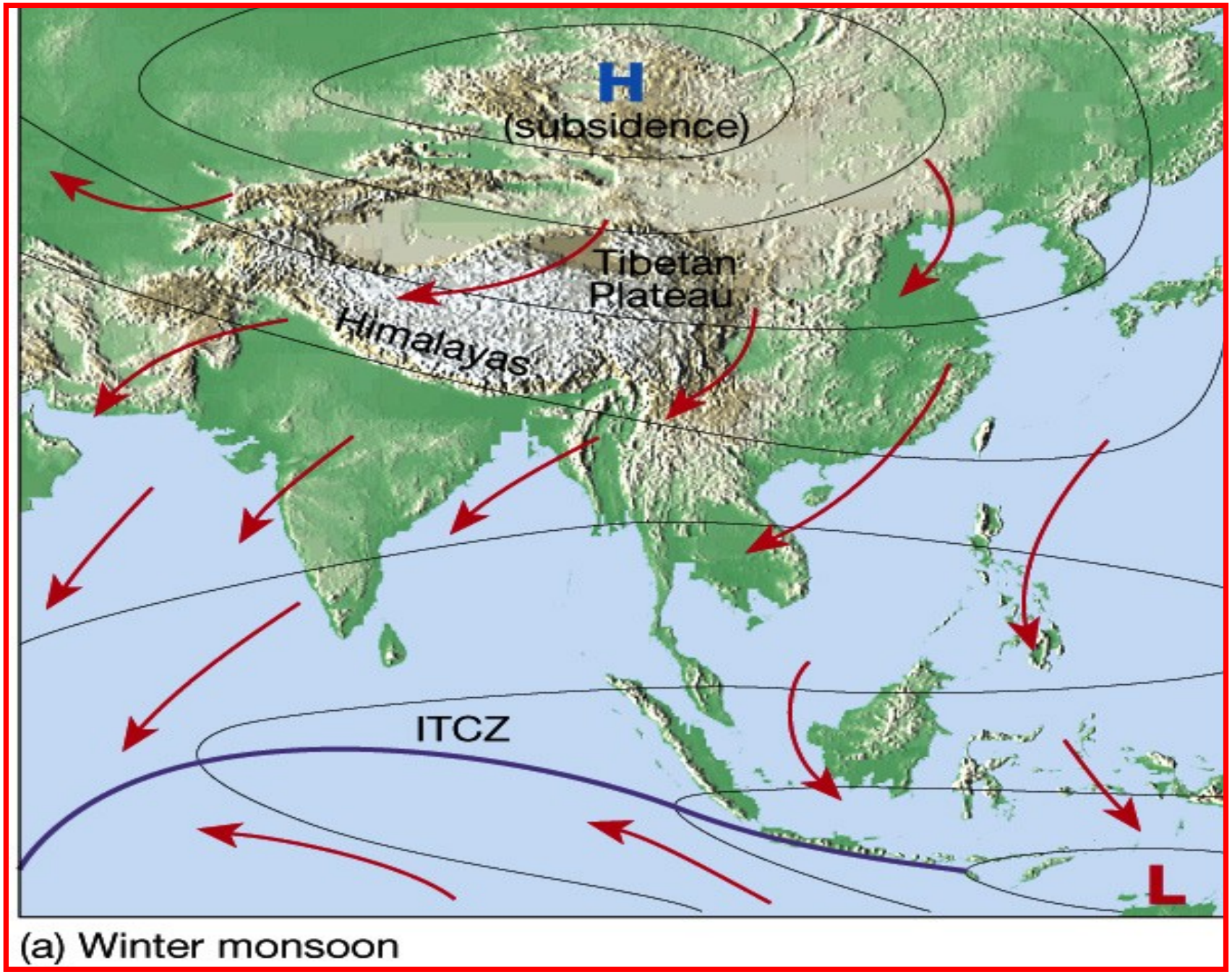
- Permanent: water temperature driven (Pacific…)
- Daily breezes: sea/lake & land; mountain & valley;
- Annual: Monsoons (India, Madagascar, SW U.S.…)


Icelandic Low
Aleutian Low
Siberian High
Azores High
Pacific High
ITCZ
ITCZ
(a) January
120°
60°
0°
60°
120°
30°
0°
30°
60°

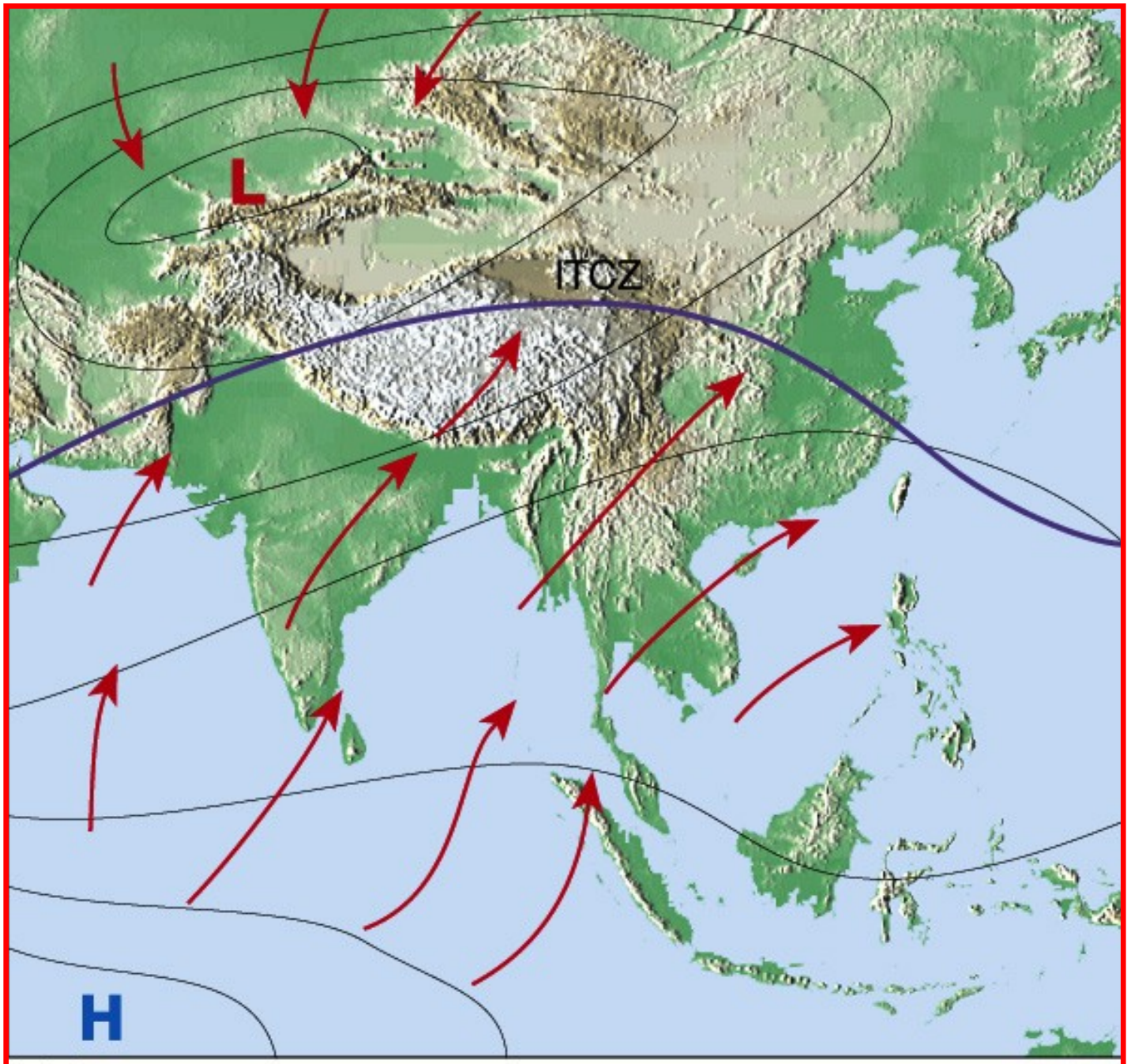

Pacific High
Bermuda High
ITCZ
ITCZ
(b) July
120°
60°
0°
60°
120°
30°
30°
60°

Monsoons

- The greatest seasonal change in Earth's global circulation is the development of **monsoons**, wind systems that exhibit a pronounced seasonal reversal in direction.
- The best-known and most pronounced monsoonal circulation, the **Asian monsoon** found in southern and southeastern Asia, is a complex seasonal change that is strongly influenced by the amount of solar heating received by the vast Asian continent.



(a) Winter monsoon



(b) Summer monsoon

North American Monsoons

- The **North American monsoon** (also called the **Arizona monsoon** and **Southwest monsoon**), a relatively small seasonal wind shift, produces a dry spring followed by a comparatively rainy summer that impacts large areas of southwestern United States and northwestern Mexico.


Southwest or Arizona Monsoon
PACIFIC OCEAN
L
Gulf of California
Moist air
Gulf of Mexico
Moist air

29Jan 2025: No Rain in Las Vegas…



The Monsoons didn’t arrive for 3 million people

Heat Records Broken, Rain Never Came: Las Vegas's 2026 Monsoon Season Has Failed

Subtropical high setting three daily records in a row also shut off Nevada's seasonal rain

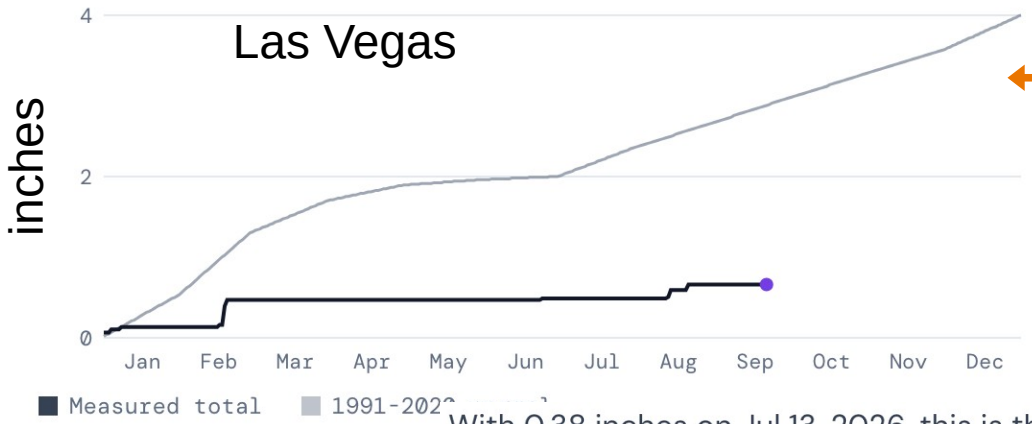
By Earl Bensen

Published: Aug 31 2026, 5:00 AM EDT



The Monsoons didn't arrive for 3 million people

With 0.24 inches on Feb 18, 2026, this is the 404th wettest of the 1,978 measurable rain days in the Las Vegas Harry Reid International Airport record, which begins in 1948.

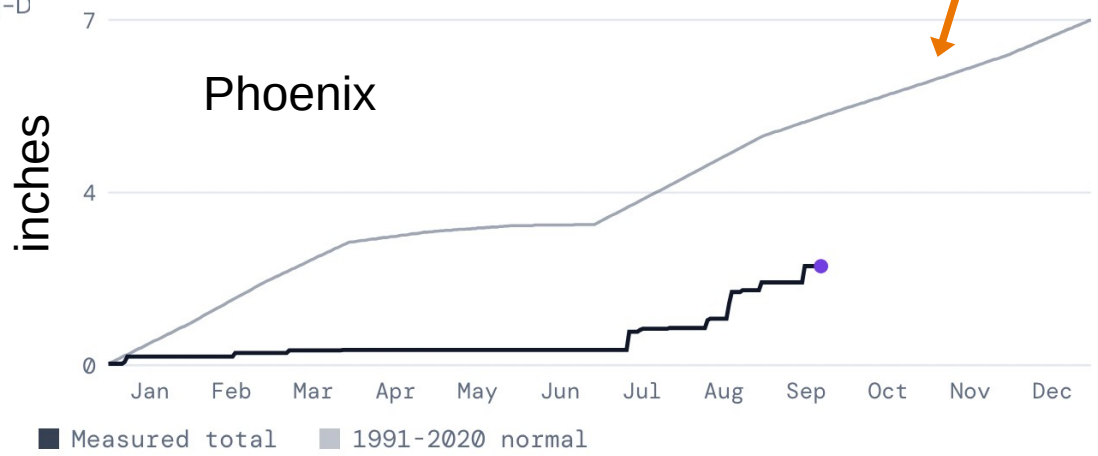


Normal

Running rain total for 2026, January
the 1991-2020 normal, violet dot m

Through Sep 21, 2026, the GHCN-D

With 0.38 inches on Jul 13, 2026, this is the 508th wettest of the 3,052 measurable rain days in the Phoenix Airport record, which begins in 1933.



Running rain total for 2026, January to December. Dark line is measured, light line is the 1991-2020 normal, violet dot marks the latest day.

Through Sep 22, 2026, the GHCN-Daily station record shows 2.07 inches of measurable rain so far this

The Monsoons didn’t arrive

Haboobs





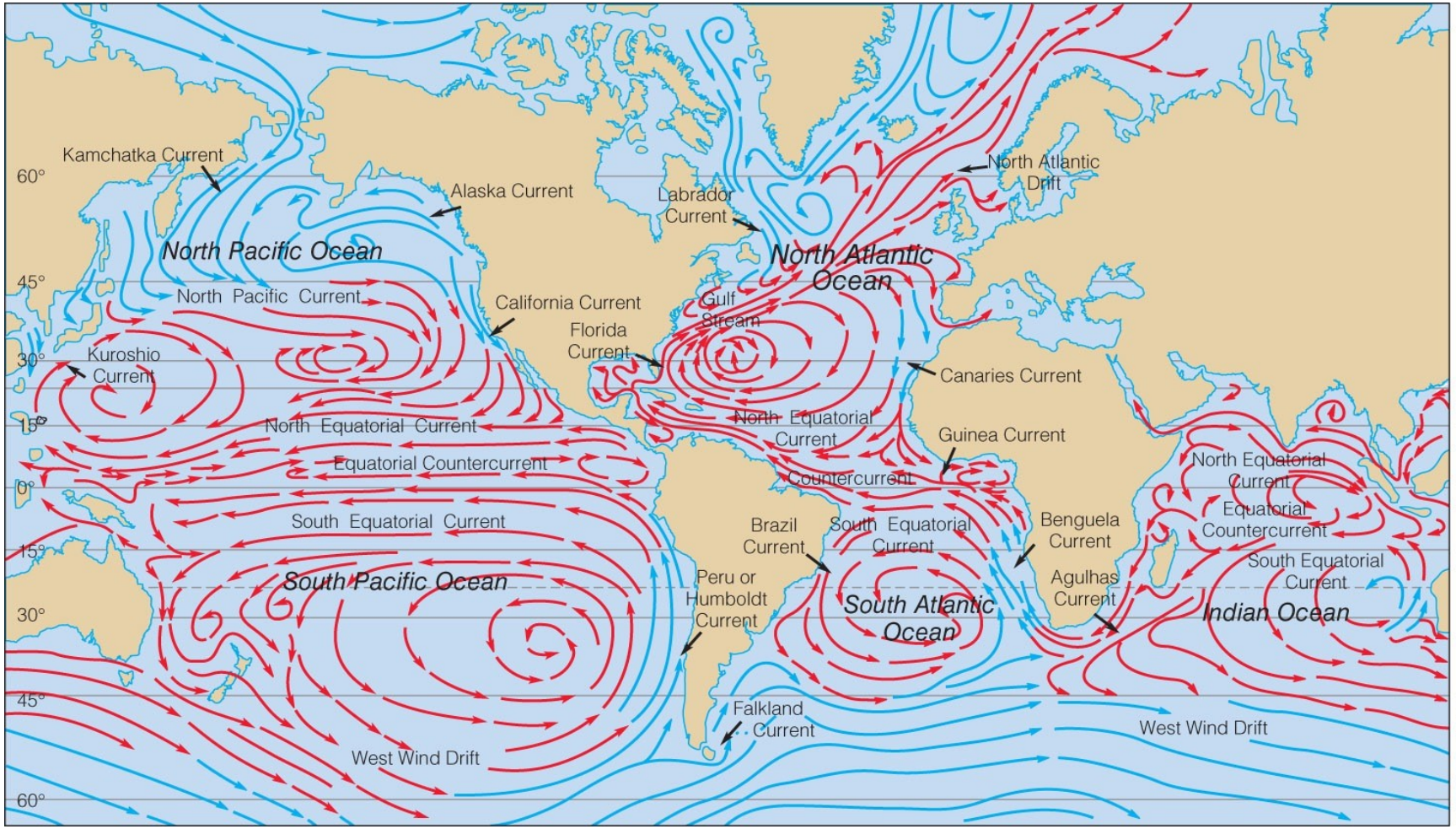
Outflow from distant thunderstorms: density currents

2026: Flooding, and now a Hurricane



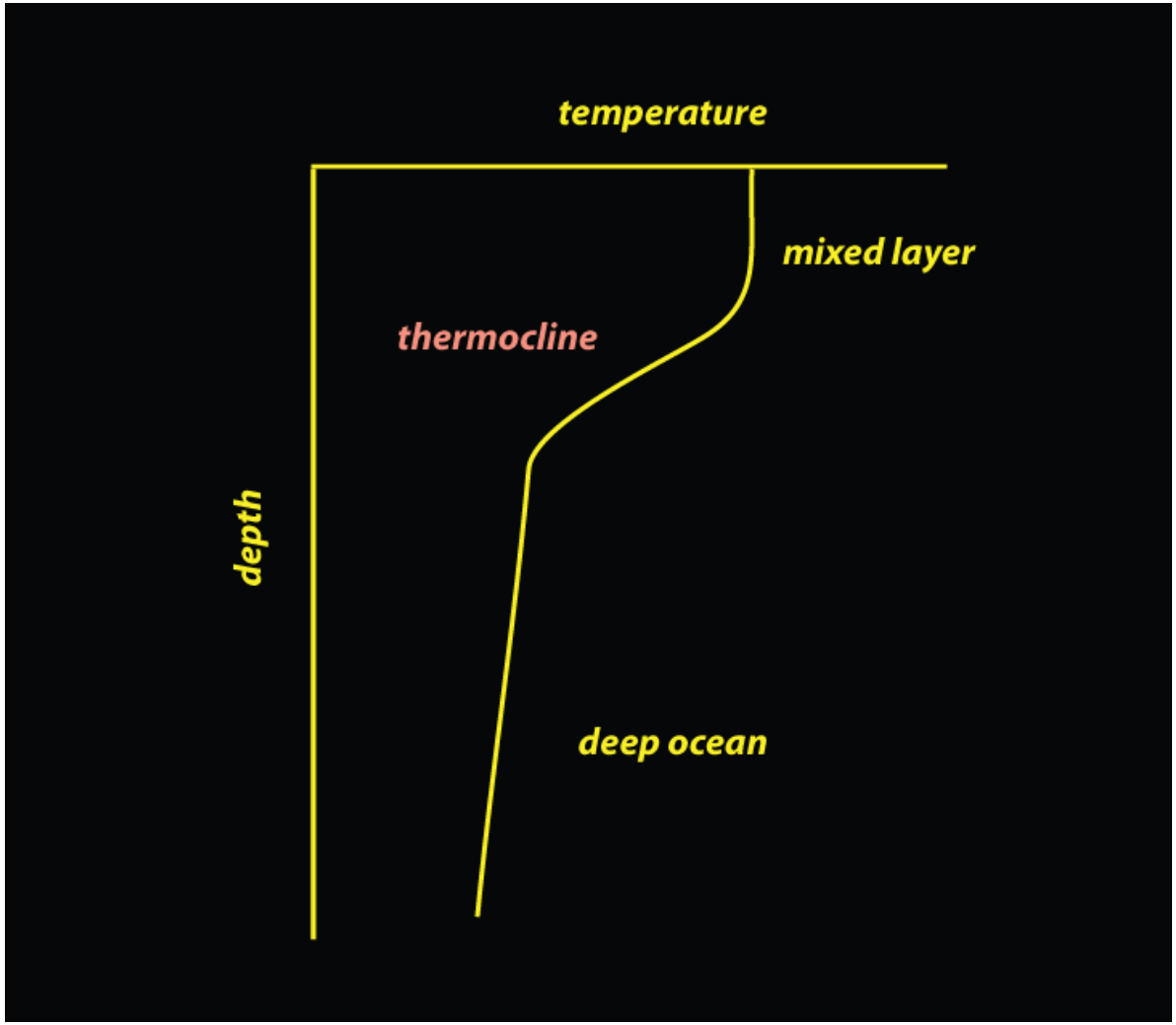
Strong Monsoon

Surface Water Circulation



Gyres are formed through winds and friction, Earth's rotation, with the location of land masses blocking the circulation

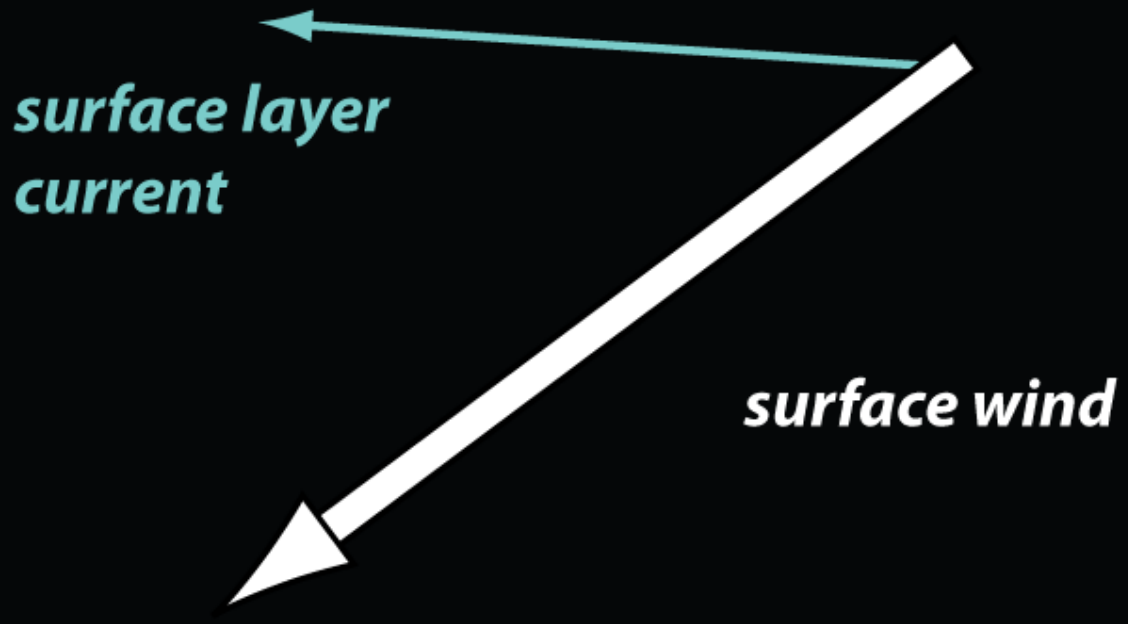
Deeper Ocean Circulation and the Polar Areas



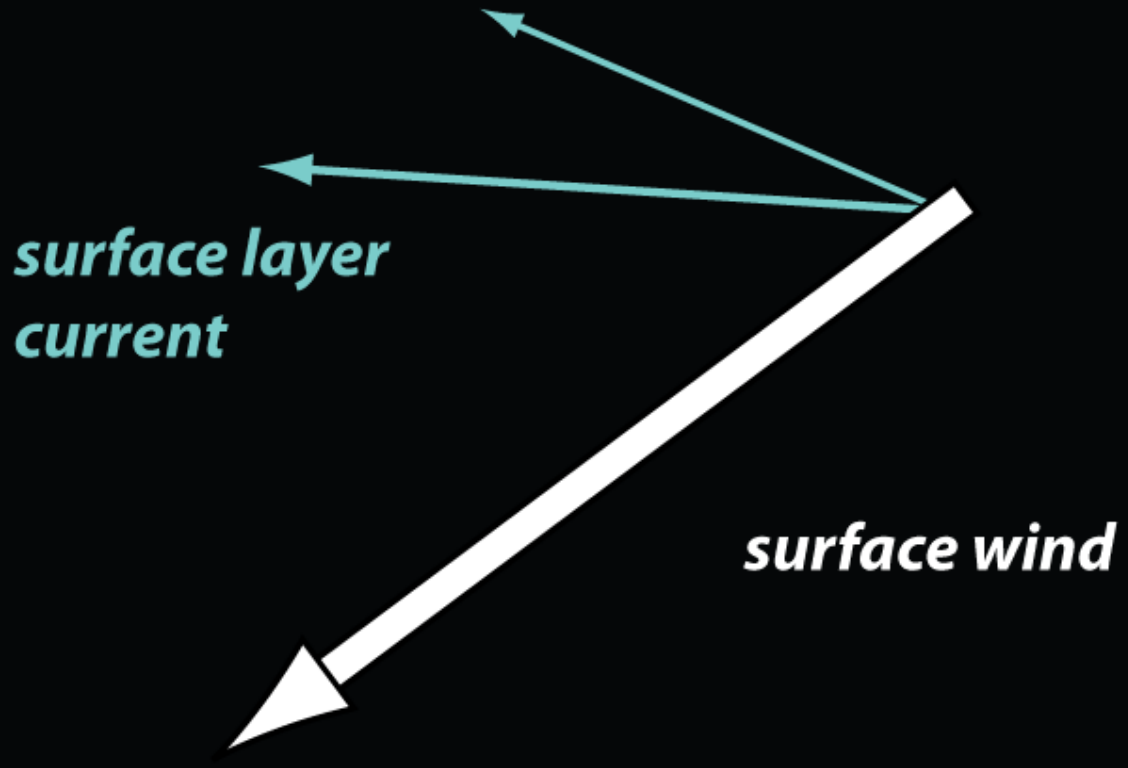
Vertical variation of temperature in the ocean



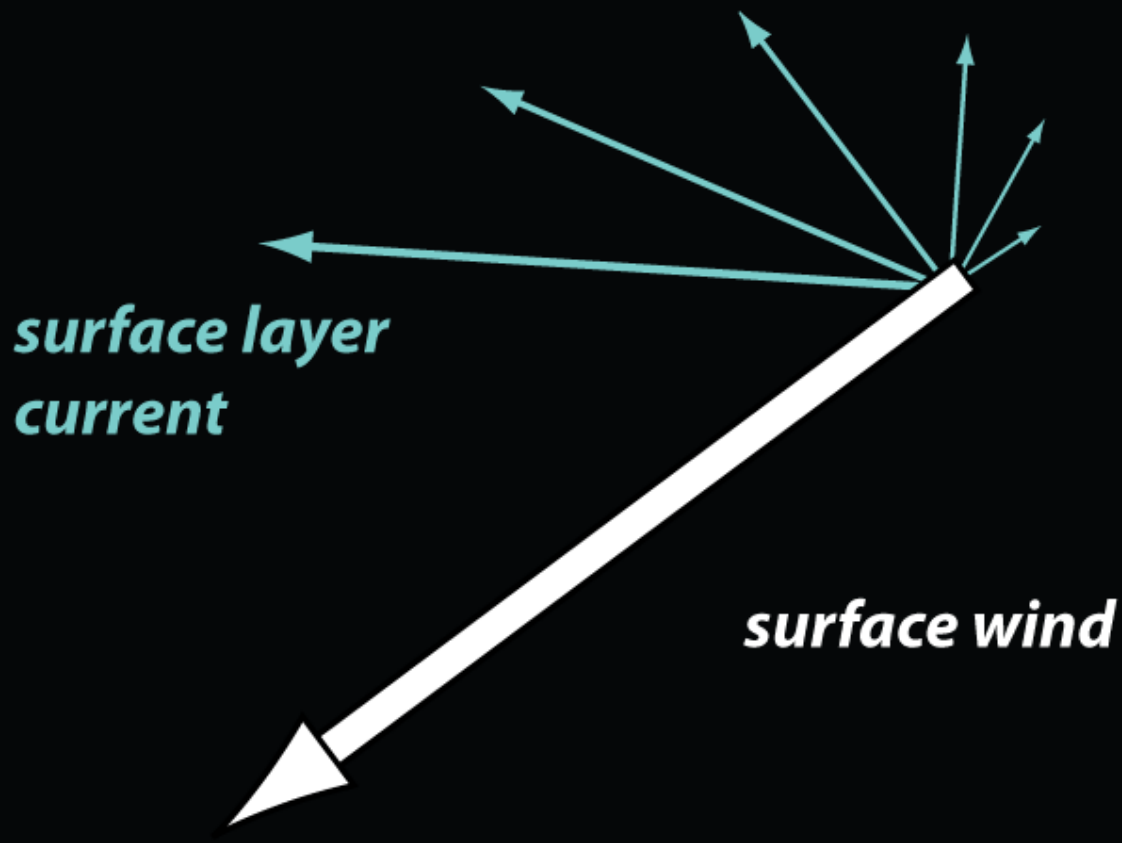
We start with a surface wind, such the NE trade winds of the tropical N Pacific.



While the wind is pushing the water SW-ward, the ocean's surface layer current is moving both more slowly, and to the RIGHT of the wind, in the NH. This is because of the Coriolis effect.

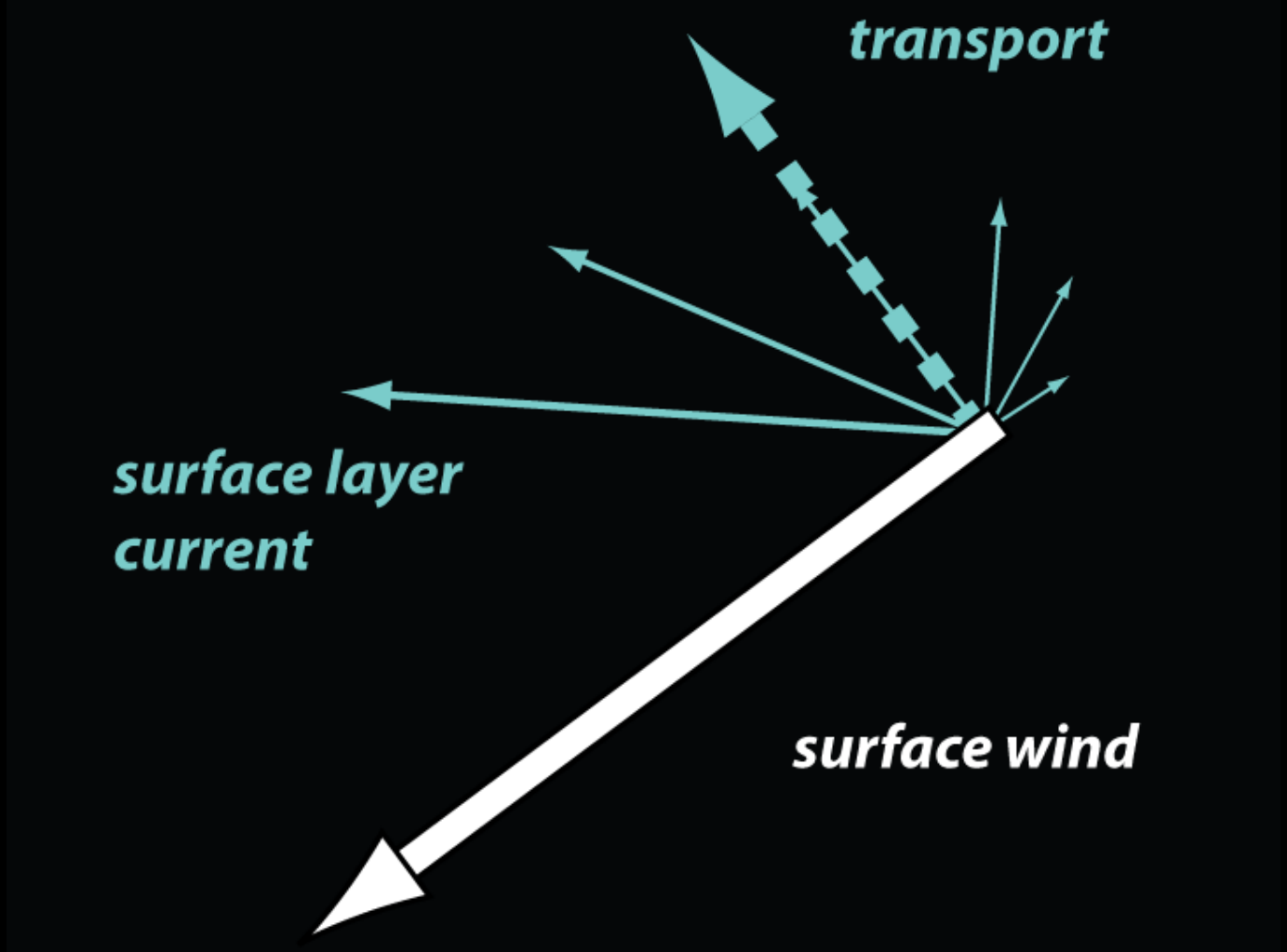


The surface layer current is dragging along water underneath, representing the primary forcing for the next layer down. But Coriolis also tries to make THAT water turn to the right of THIS forcing. So at this depth, the current is weaker, but even MORE turned rightward relative to the surface wind.



This keeps going, with the current both slowing and turning CW with increasing depth. By about 100 m down, the water is flowing in the OPPOSITE direction to the surface wind!

This is called an EKMAN SPIRAL.



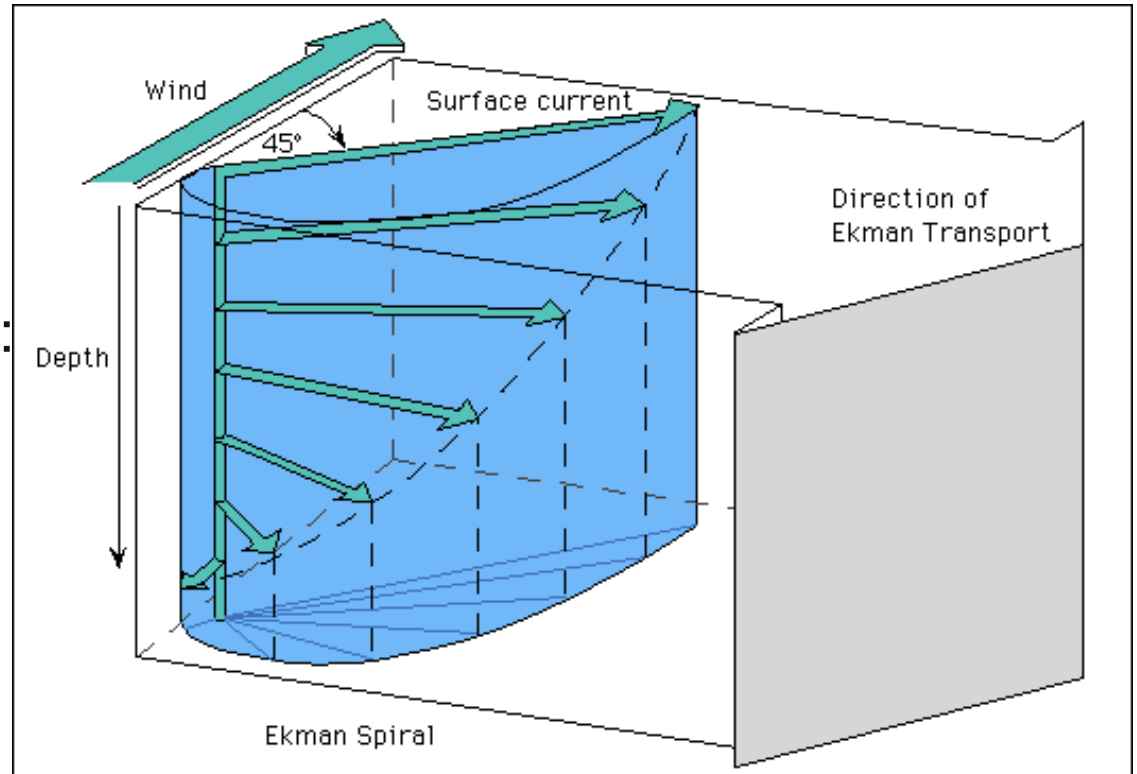
The Ekman spiral has an important consequence. The layers drawn don't represent an equal amount of water, but Average Direction is about 90 degrees to the RIGHT of the surface wind direction.

This mass motion is the EKMAN TRANSPORT.

Ekman velocity spiral

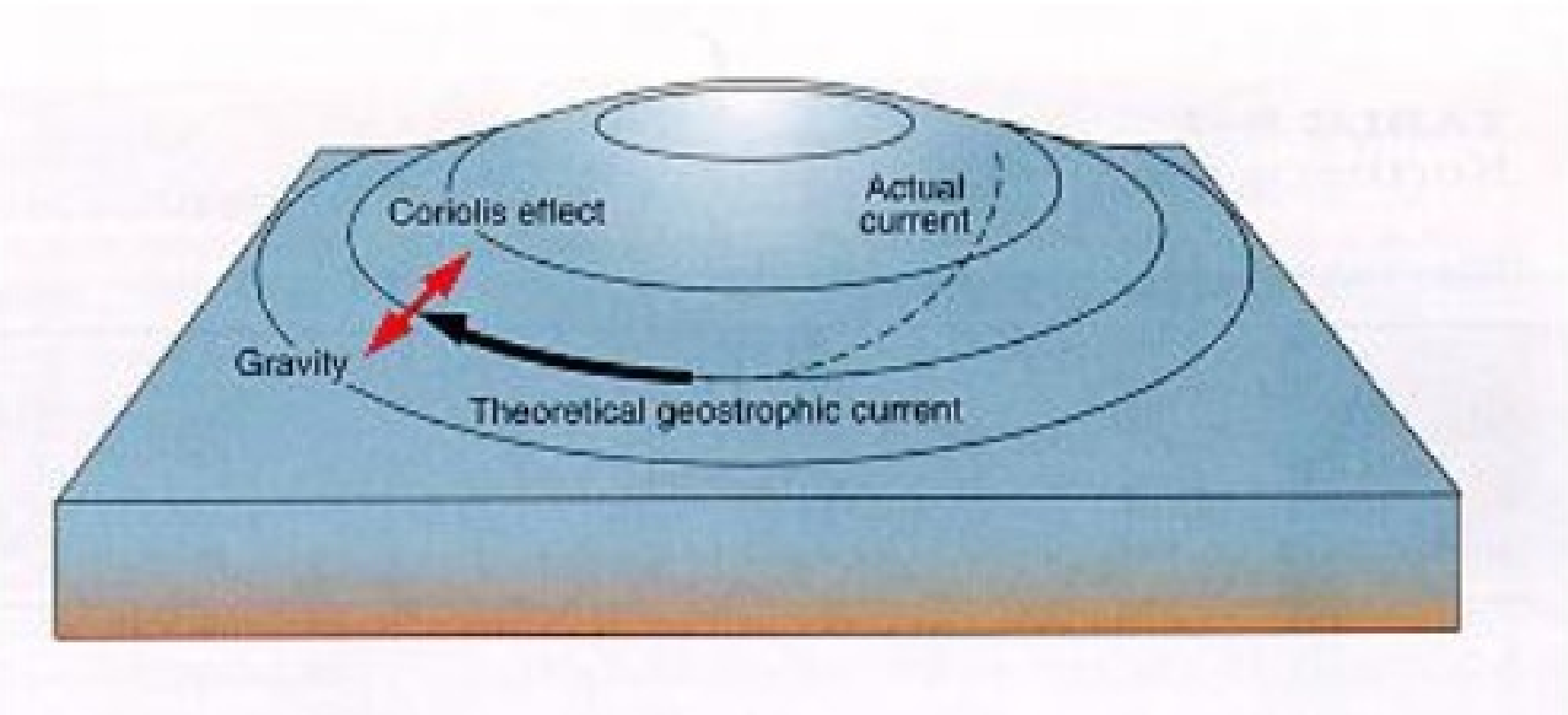
- Surface velocity to the right of the wind (northern hemisphere, due to Coriolis)
- Surface layer pushes next layer down slightly the right, and slightly weaker current
- Next layer pushes next layer, slightly to right and slightly weaker current
- Producing a “spiral” of the current vectors, to right in northern hemisphere, decreasing speed with increasing depth

Vertical scale: about 50-100 meters



Adapted from Thurman, Harold V. **Essentials of Oceanography, 5th ed.** Prentice-Hall, Inc., 1996.

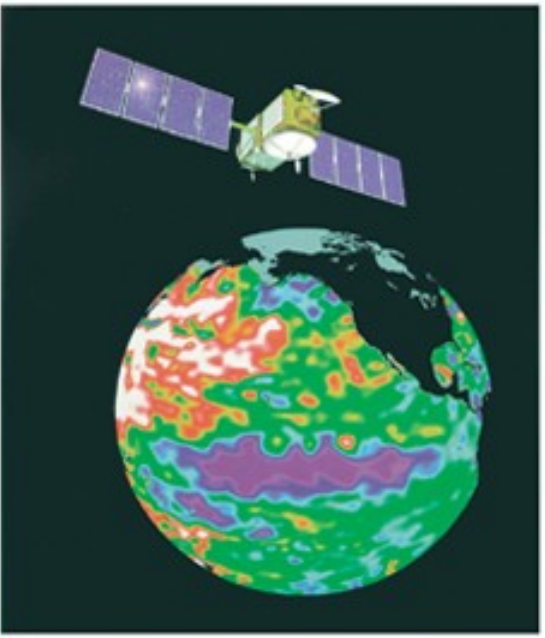
(Geostrophic) Flow: The *Hill* in the Gyre



Geostrophic Flow: The *Hill* in the Gyre

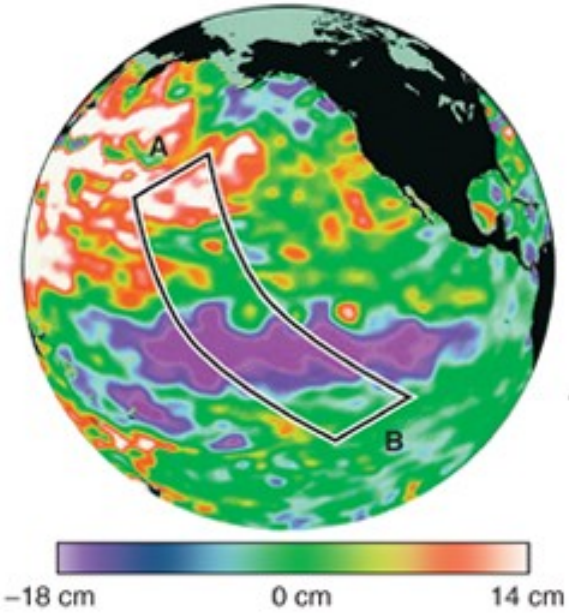
Satellite Jason 2001

32 cm difference in elevation between points A and B

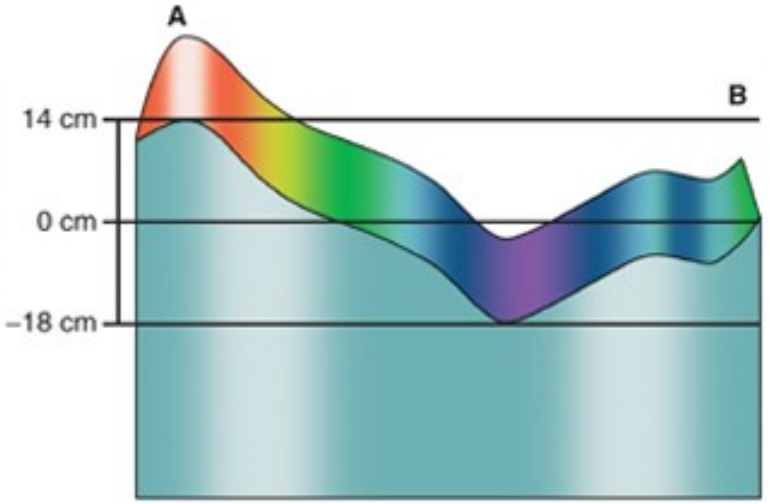


a.

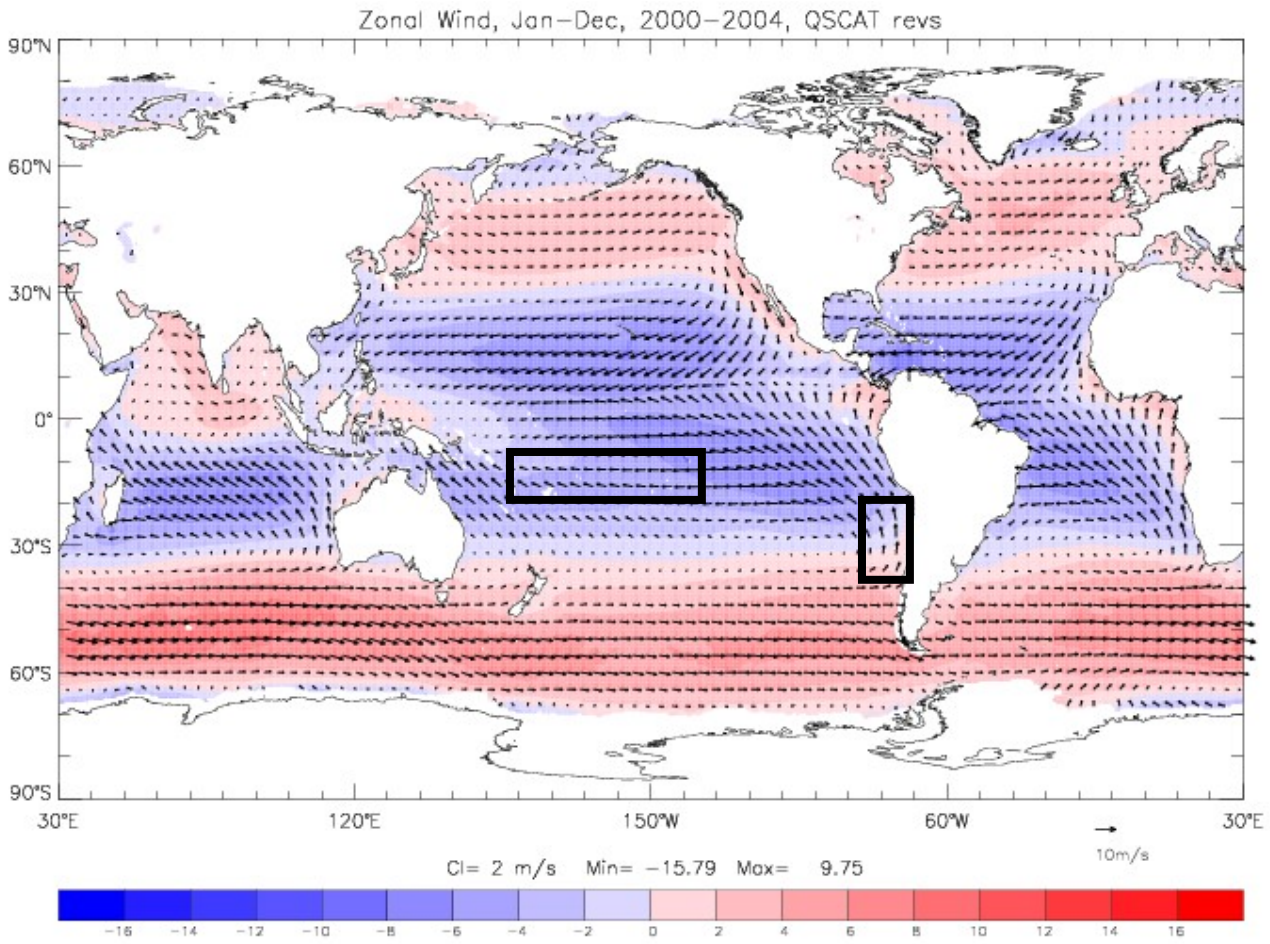
© NASA



b.



Global surface wind velocity



Westward Eastward

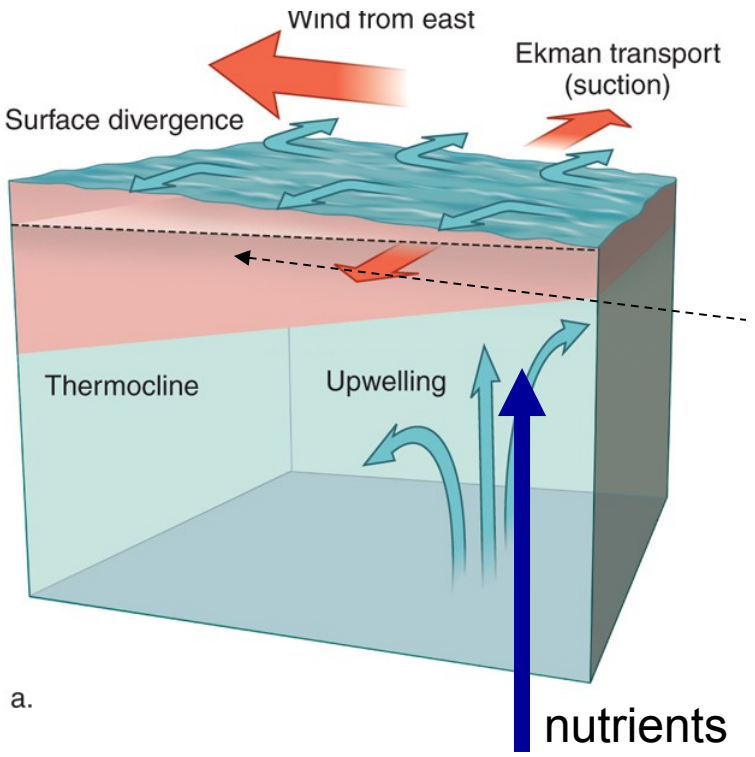
Upwelling/Downwelling

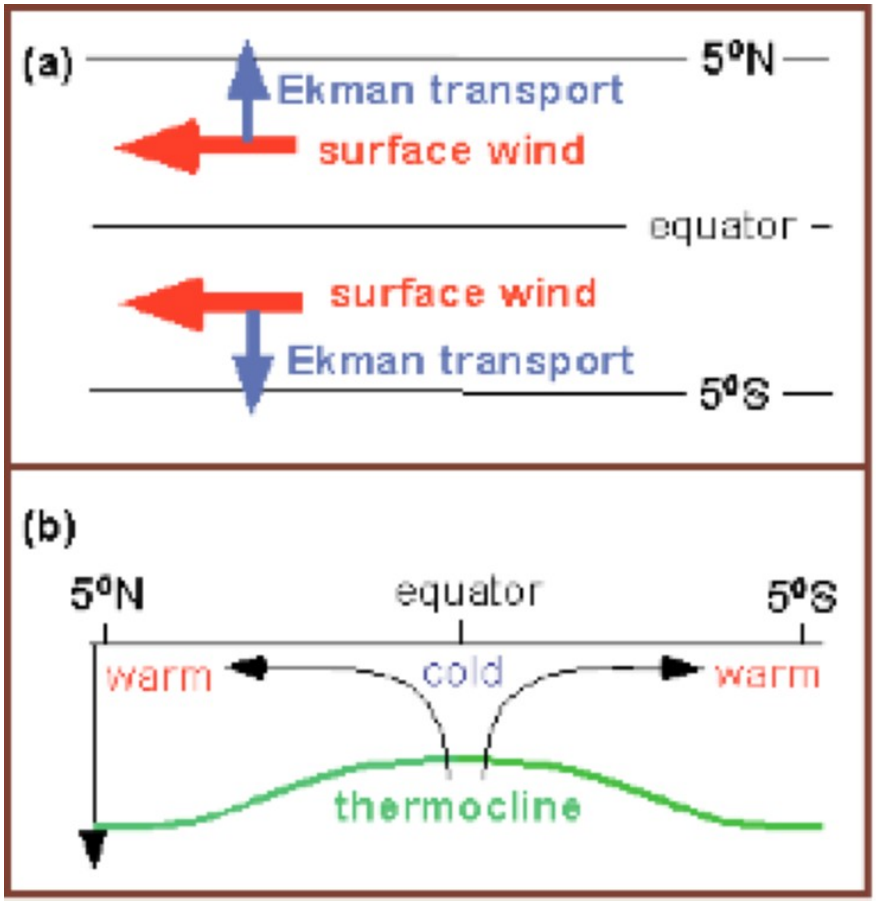
What is upwelling?

The flow of subsurface waters to the surface

Two types: Coastal and Equatorial

EQUATORIAL Upwelling

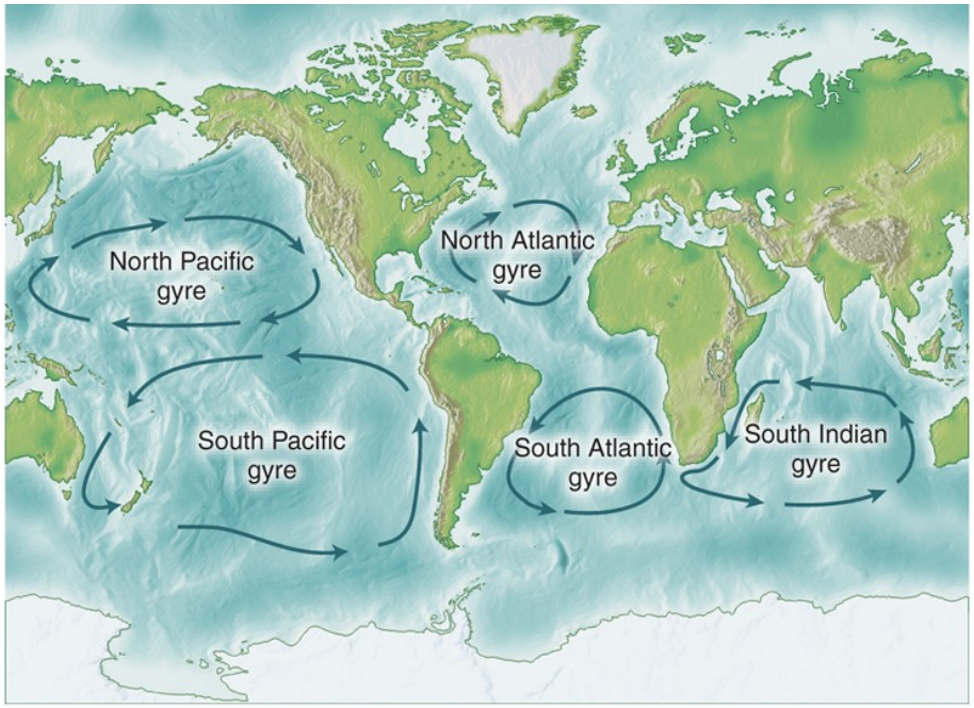


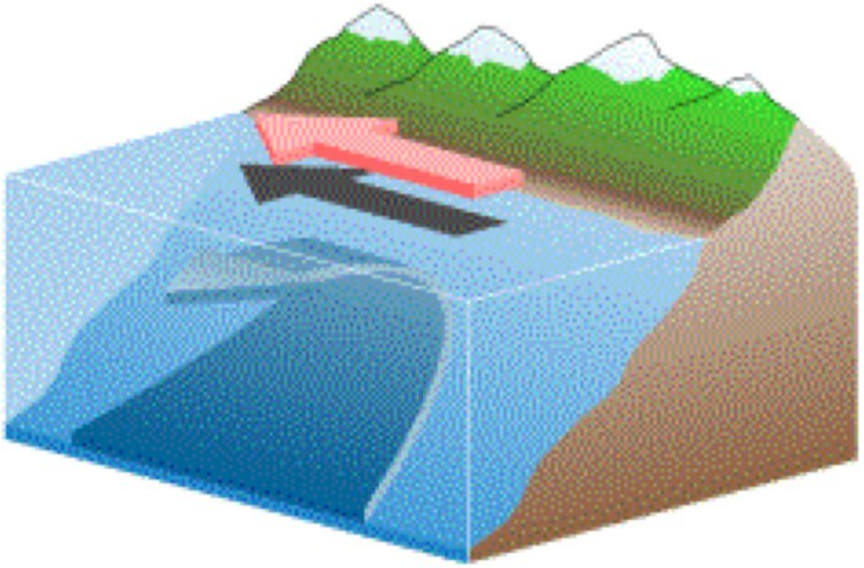


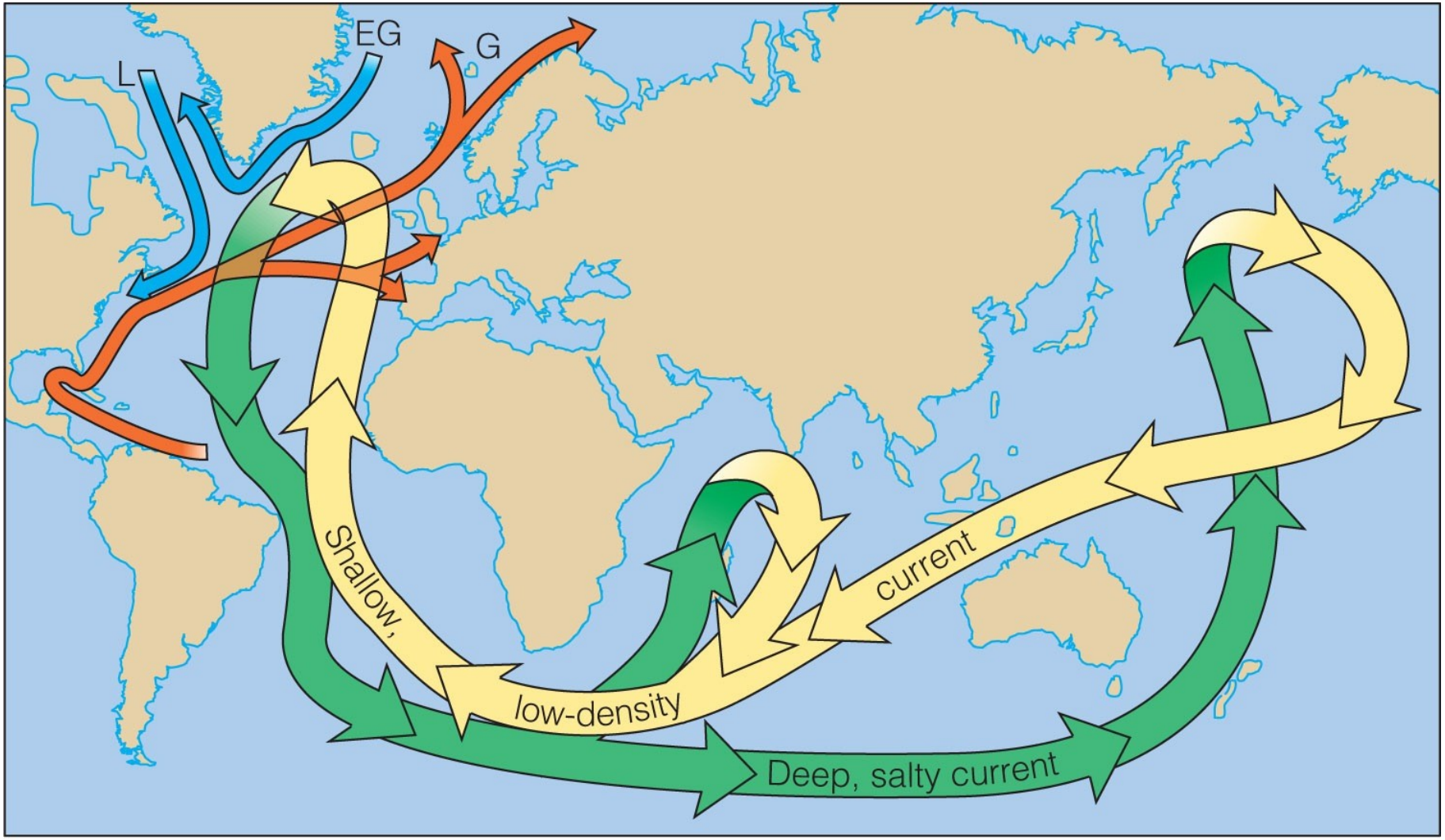
Ekman divergence (Ekman upwelling) at equator and at land boundaries

Land boundary

(southern hemisphere, like Peru)





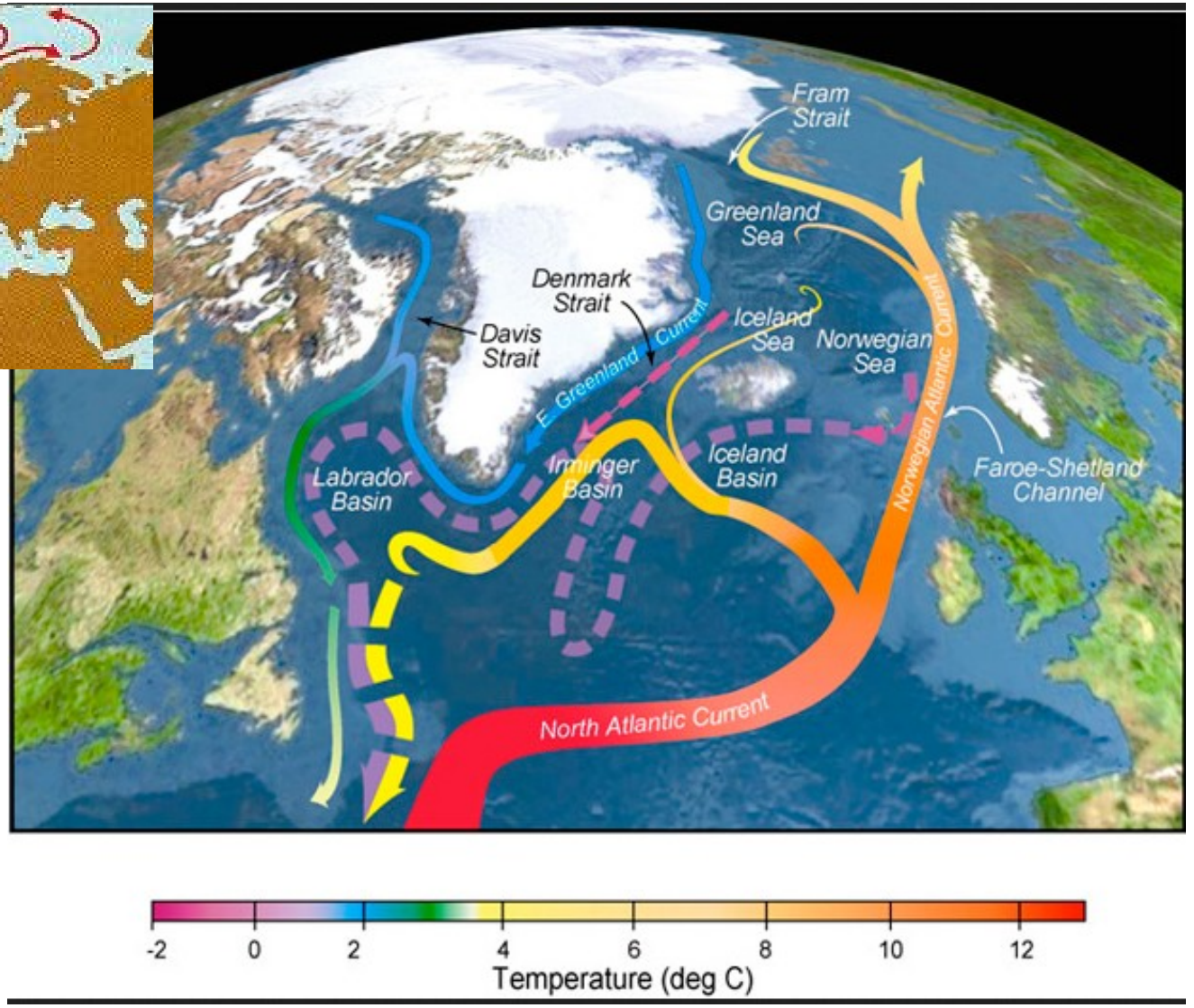


Data from Broecker, W. S. 1995, November. Chaotic climate. *Scientific American* 273[5].)

Figure 2-7 p30

North Atlantic Drift





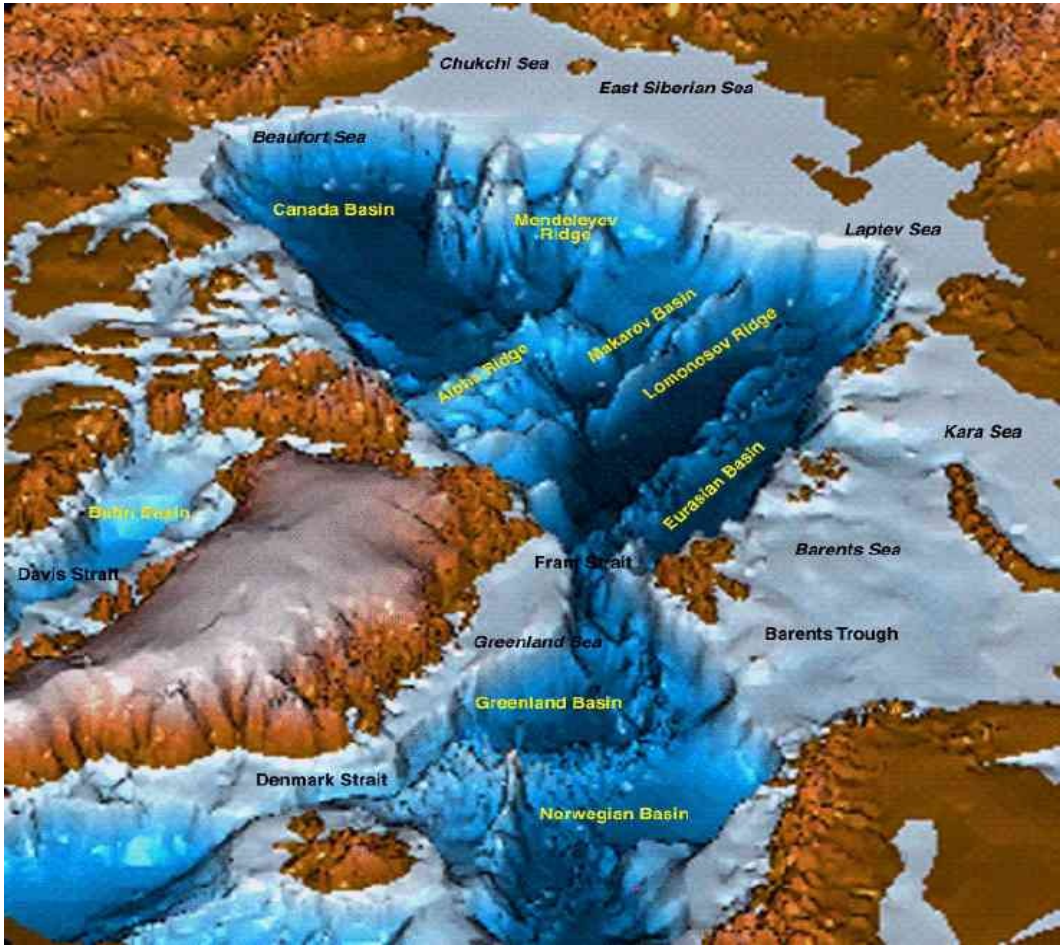
Arctic Ocean Currents



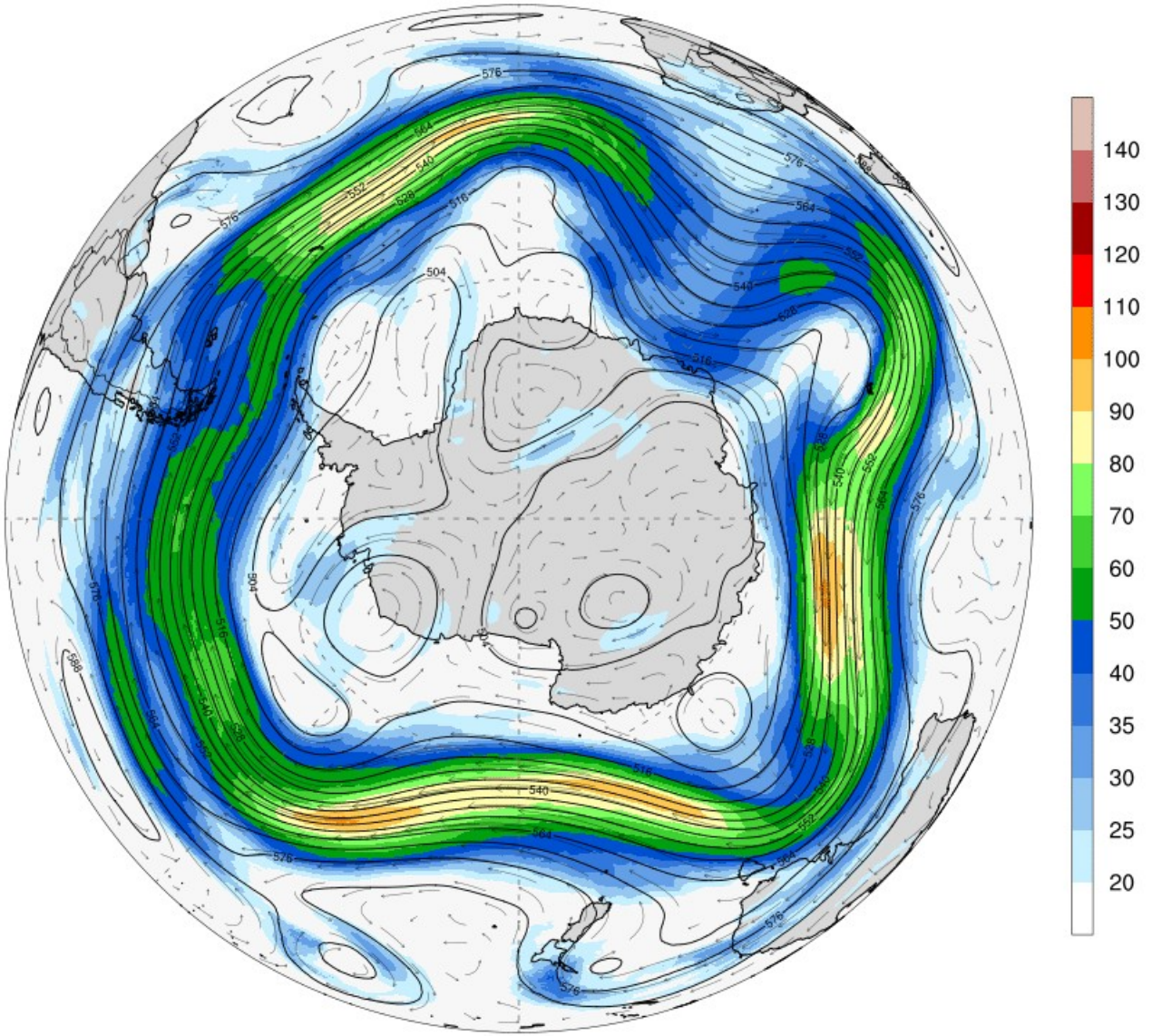
Mean Sea Ice Drift



Arctic Ocean Topography



January southern polar Jet Stream

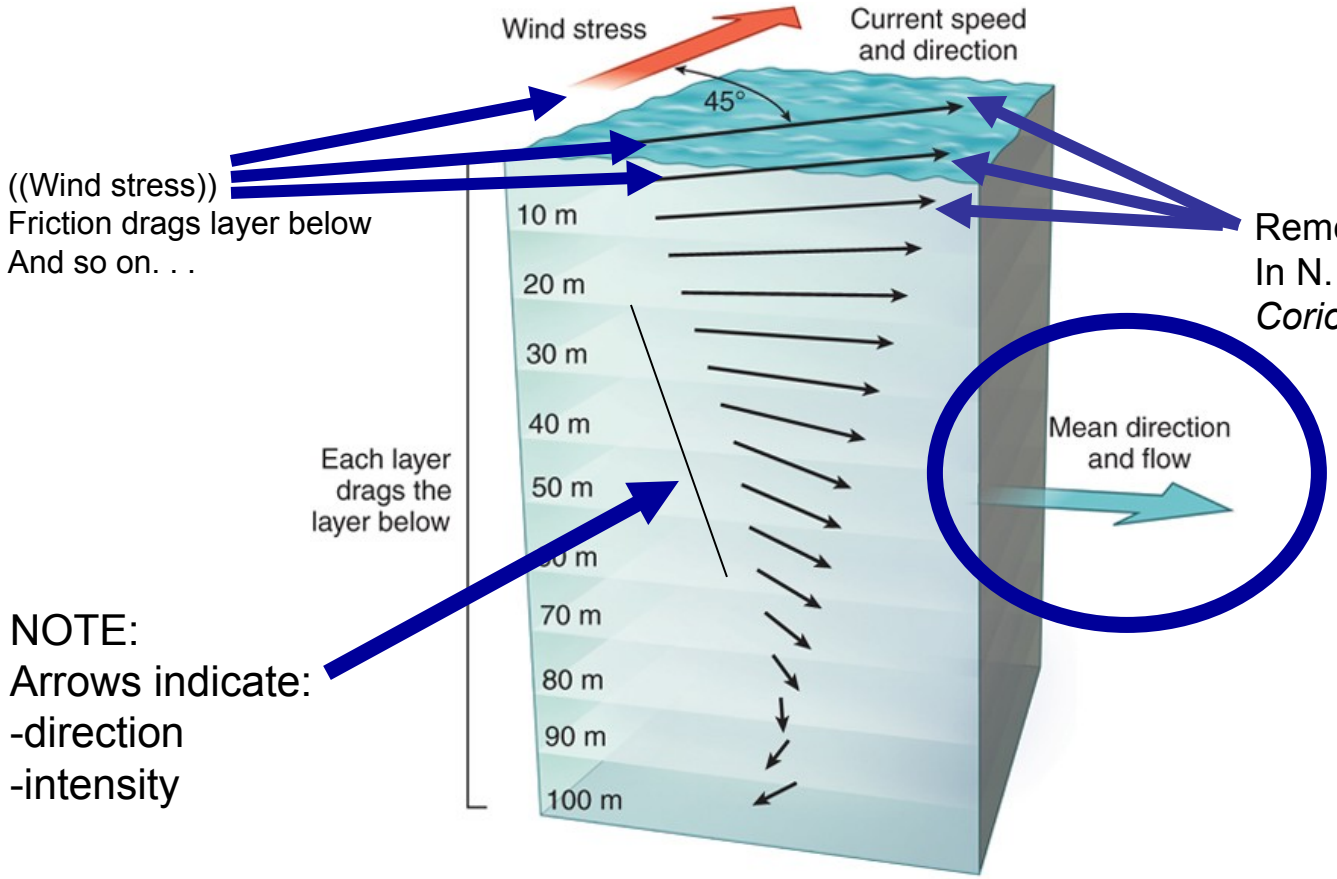


Wraps around the Antarctic continent

Motion in the Ocean The existence of the Gyres

V. Walfrid Ekman (1861-1930)

Ekman Transport



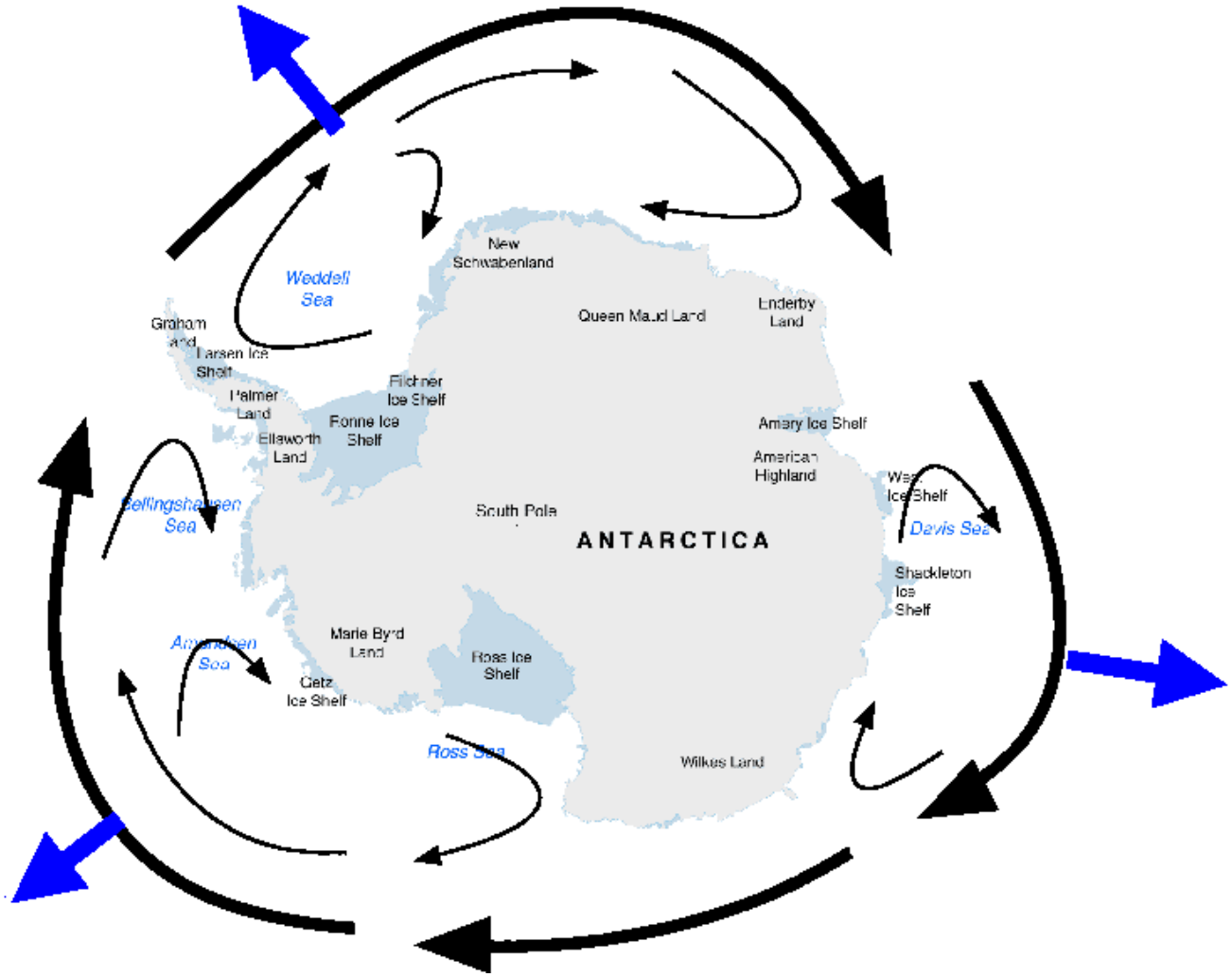
((Wind stress))
Friction drags layer below
And so on. . .

Remember:
In N. Hemisphere
Coriolis deflects this *to Right*

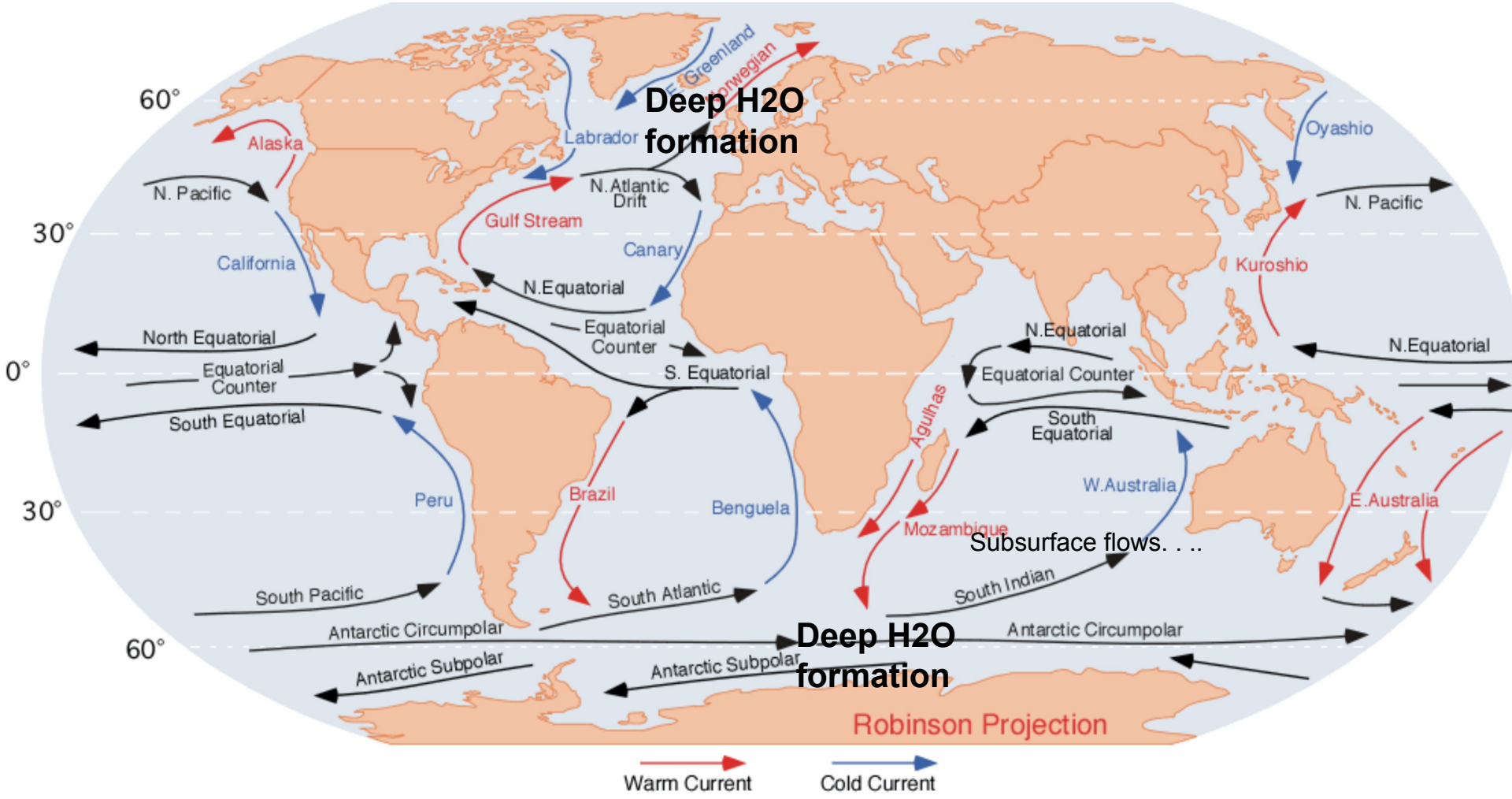
NOTE:
Arrows indicate:
-direction
-intensity

In southern hemisphere, *Coriolis* deflects water to the *Left*

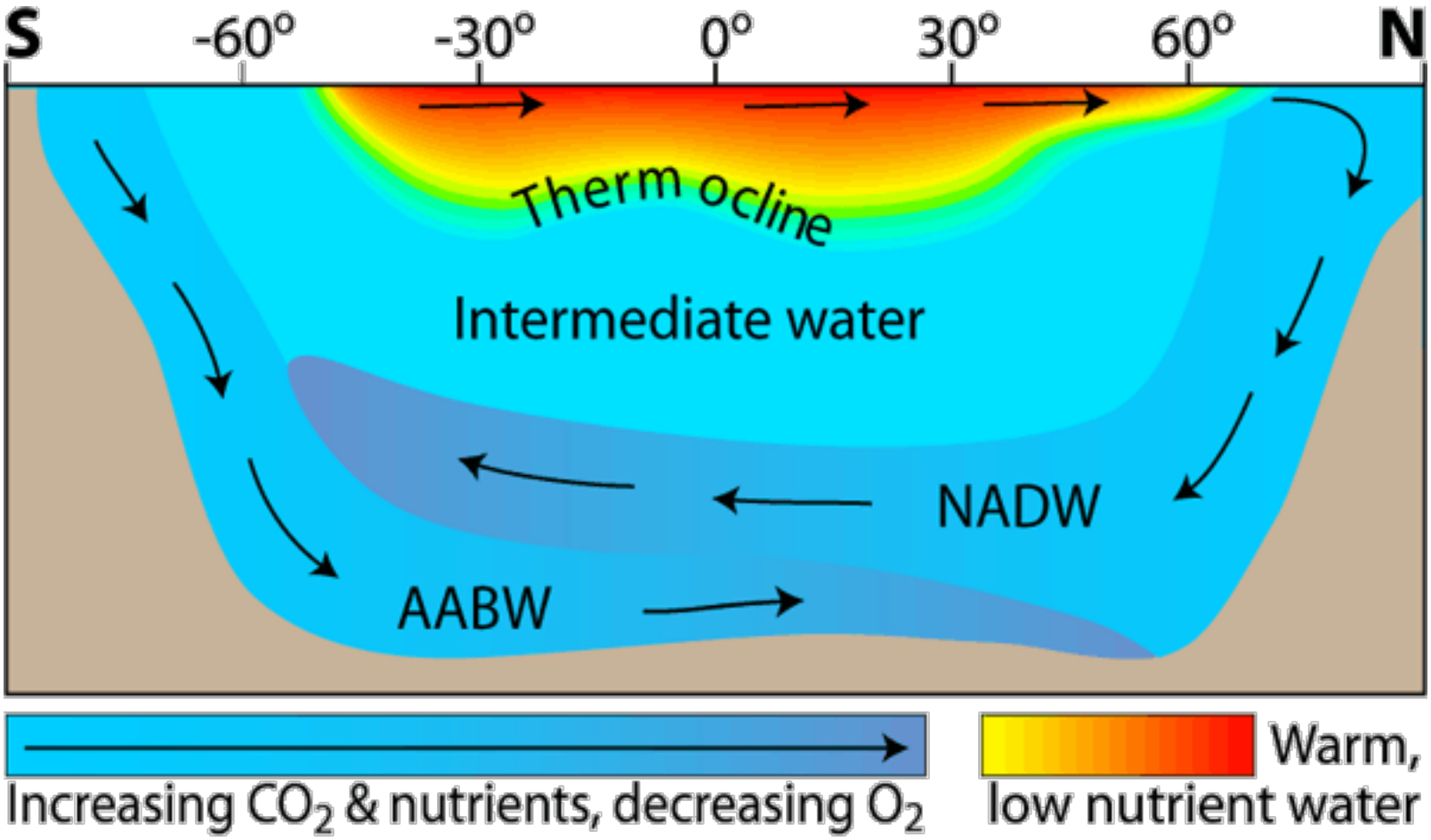
Southern Ocean



Ocean Circulation, University Press

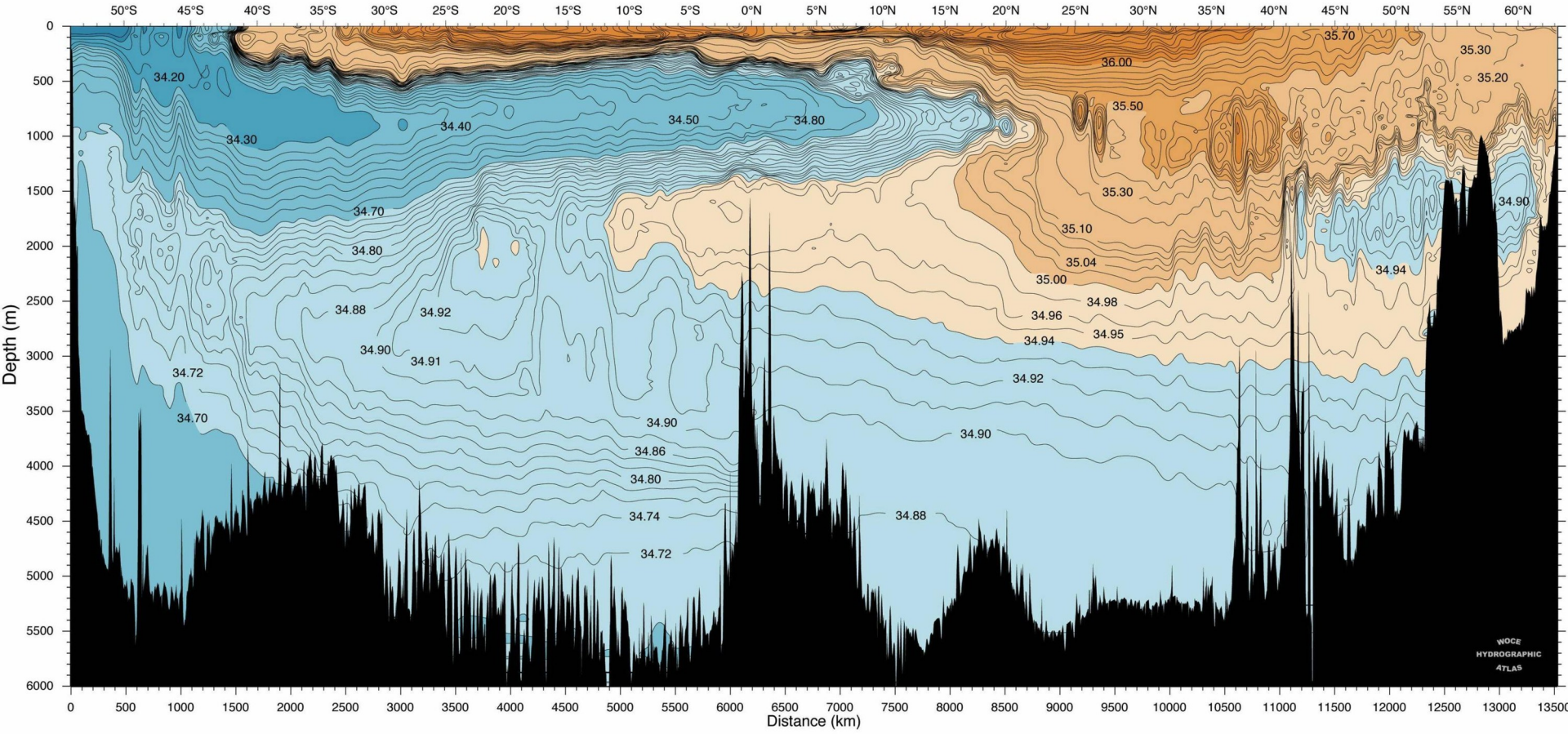

Deep H2O formation
Greenland
Norwegian
Labrador
Alaska
N. Pacific
N.Atlantic Drift
Gulf Stream
Canary
California
N.Equatorial
North Equatorial
Equatorial Counter
S. Equatorial
South Equatorial
Oyashio
N. Pacific
Kuroshio
N.Equatorial
N.Equatorial
Equatorial Counter
South Equatorial
Agulhas
W.Australia
E.Australia
Mozambique
Peru
Brazil
Benguela
Subsurface flows. . ..
South Pacific
South Atlantic
South Indian
Antarctic Circumpolar
Antarctic Circumpolar
Deep H2O formation
Antarctic Subpolar
Antarctic Subpolar
60°
30°
0°
30°
60°
Robinson Projection
Warm Current
Cold Current

Water Masses
Atlantic Sector

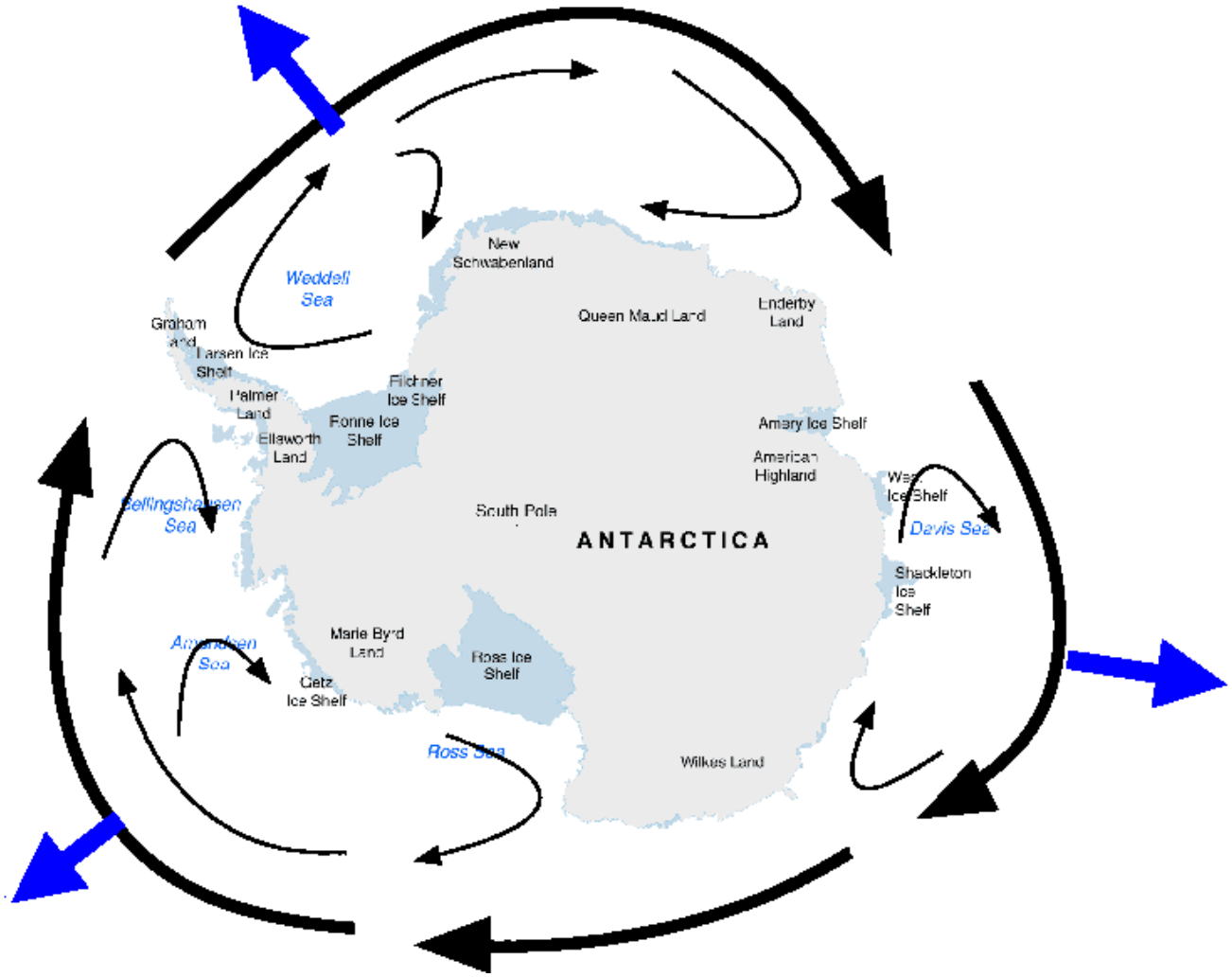


Antarctic Atlantic Bottom Water

Ocean Salinity
Atlantic Sector

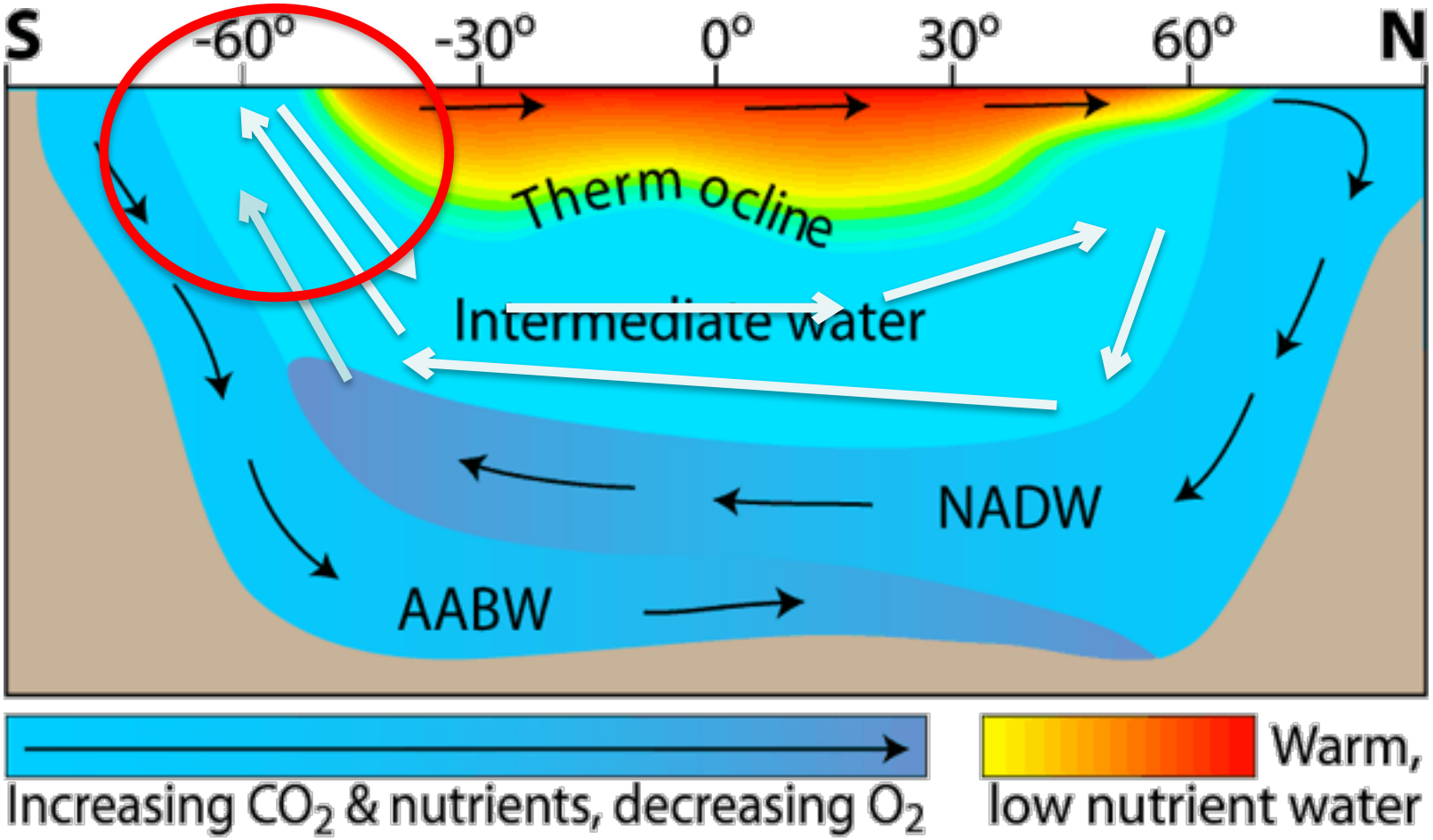


Southern Ocean

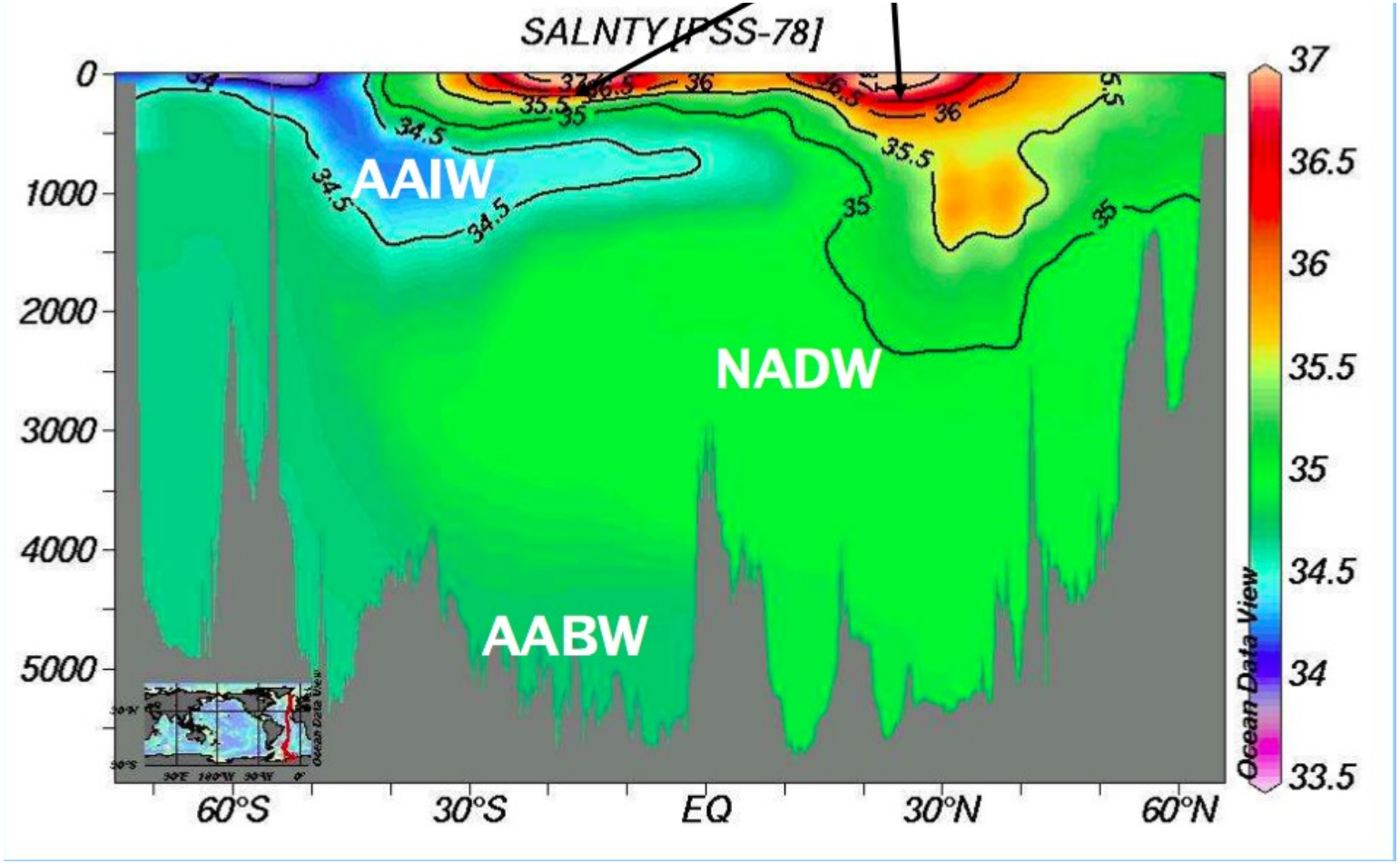


Ocean Circulation, University Press

Southern Ocean Currents
Water Masses

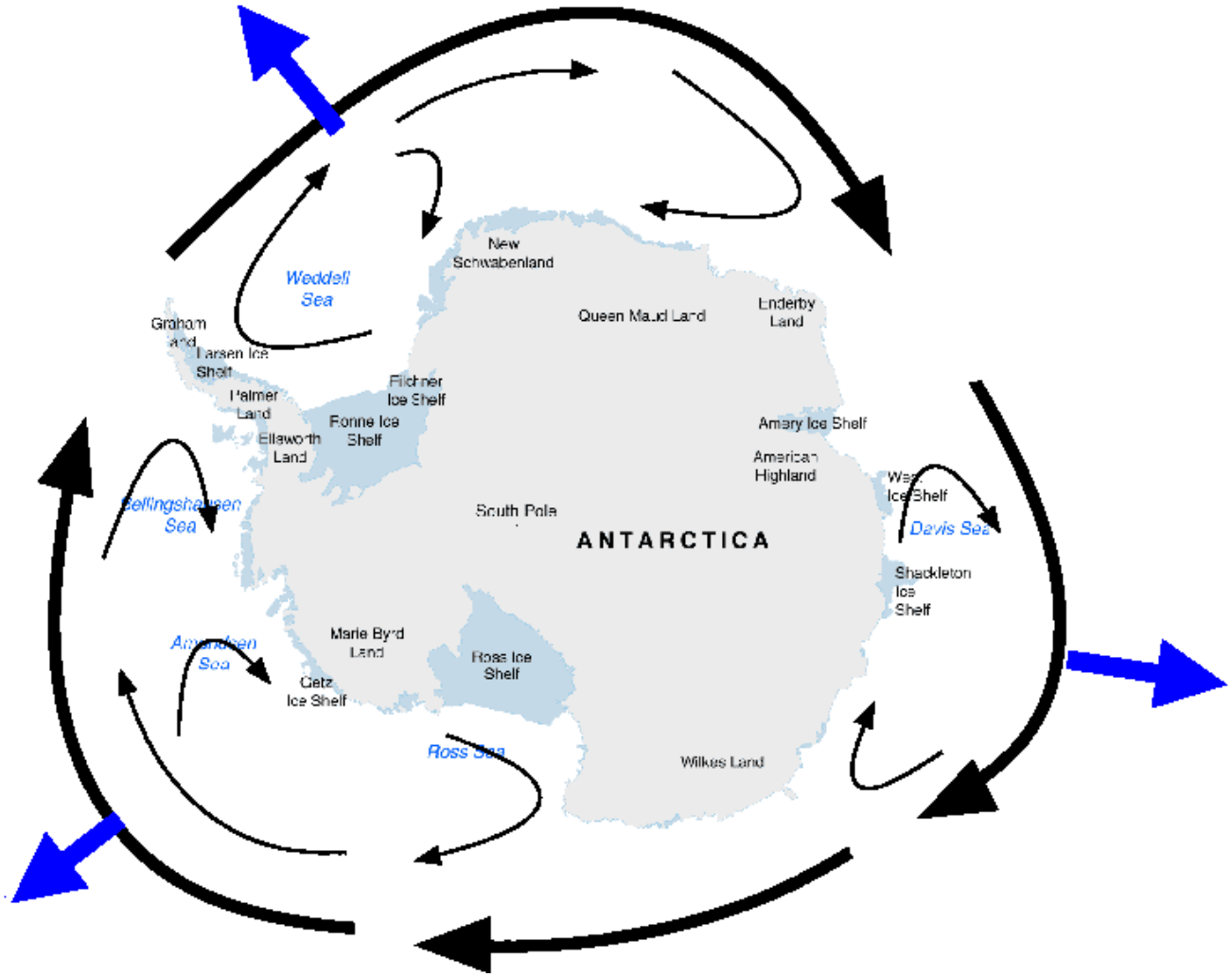


Ocean Salinity and Water Masses
Atlantic Sector

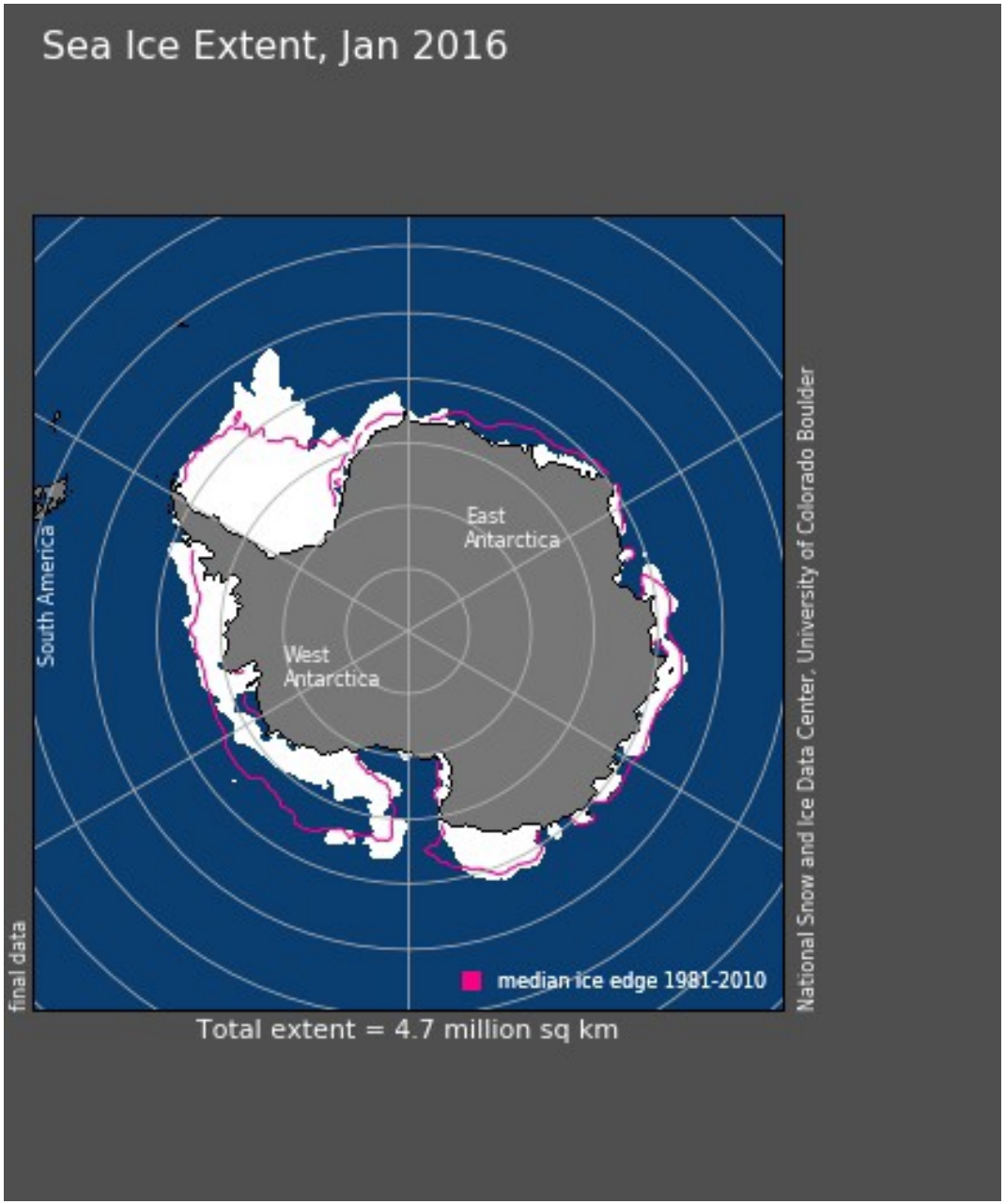


https://en.wikipedia.org/wiki/Antarctic_Intermediate_Water

Southern Ocean

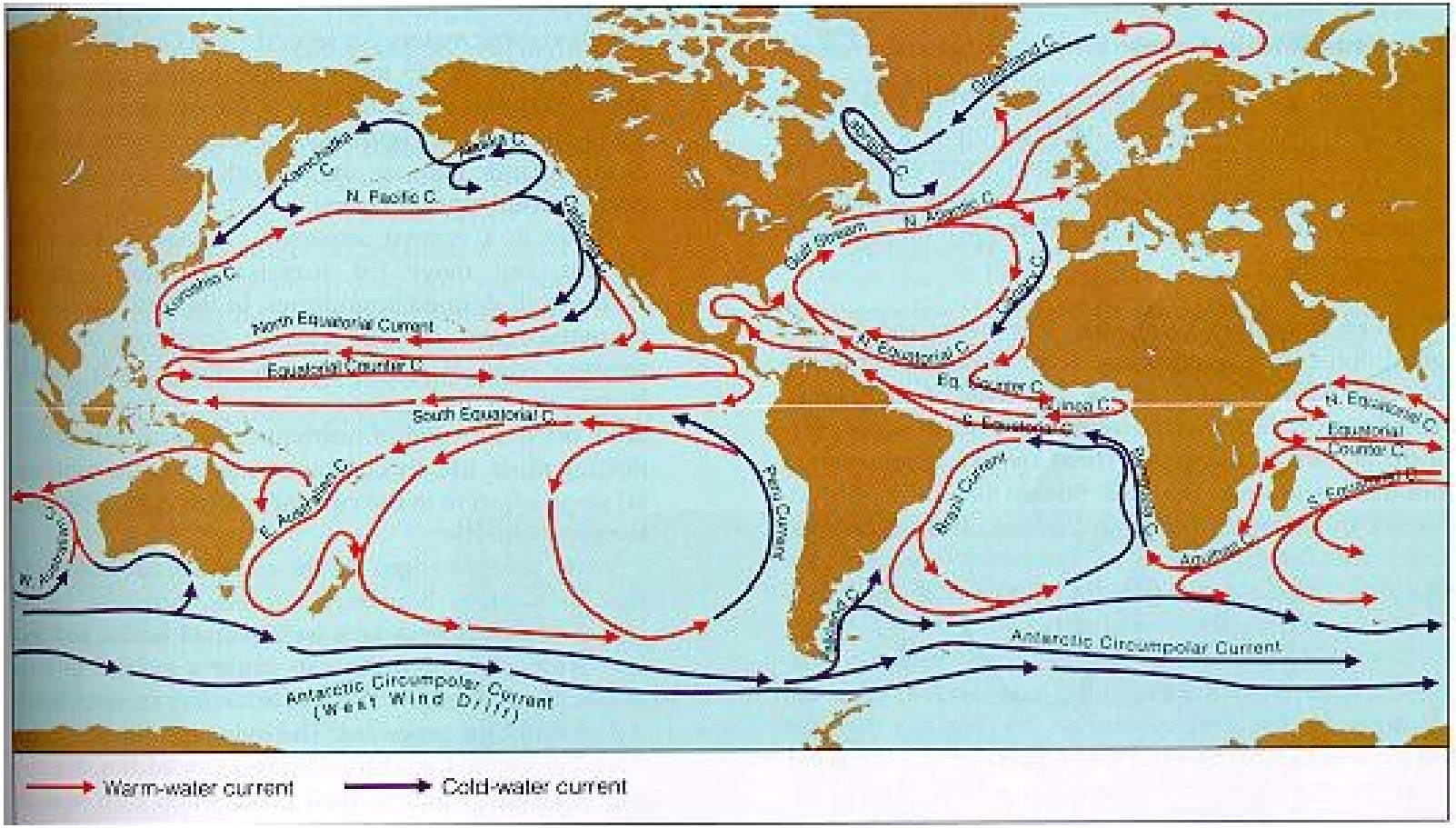


Ocean Circulation, University Press

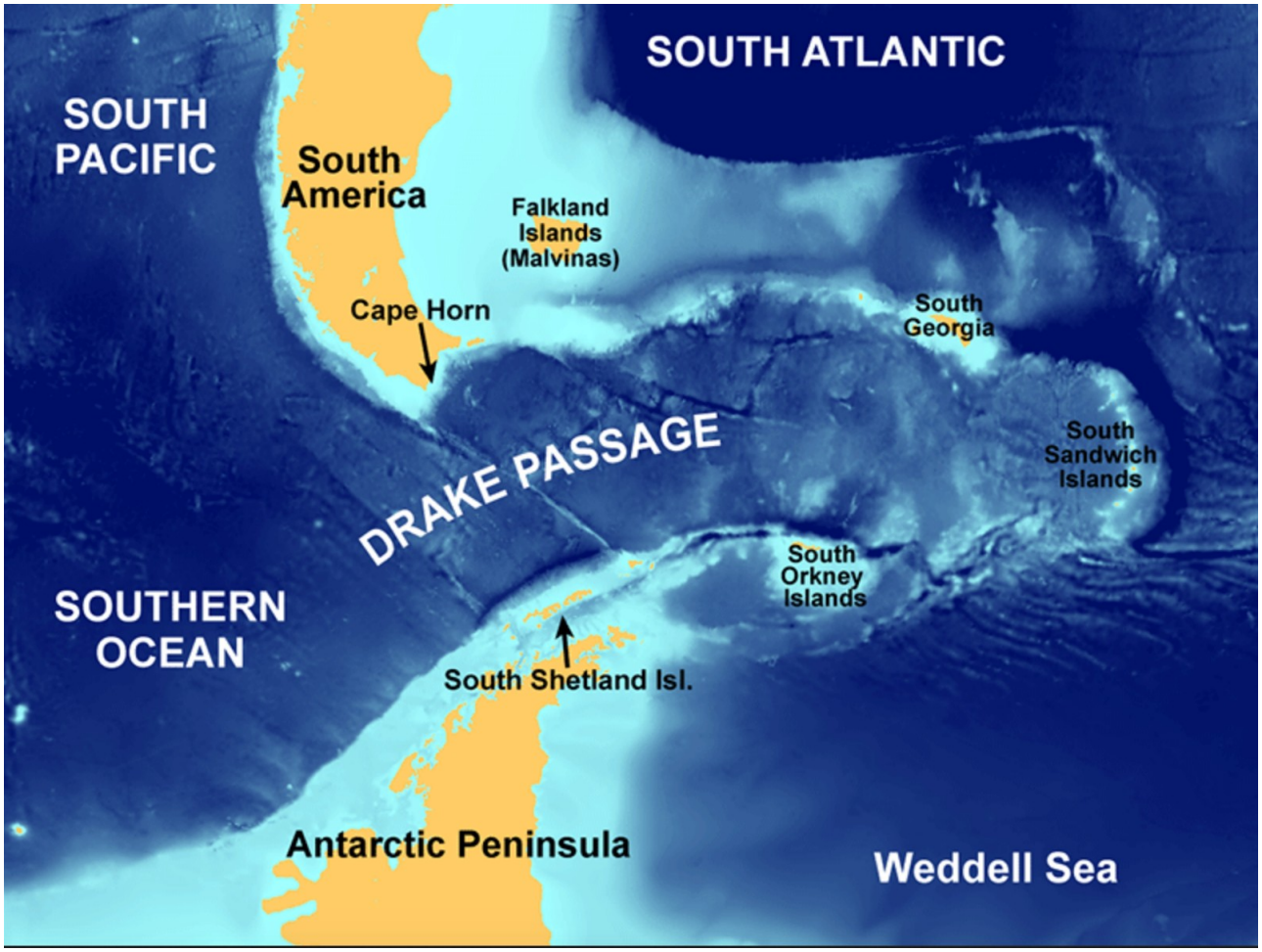

Sea Ice Extent, Jan 2016
East Antarctica
West Antarctica
South America
median ice edge 1981-2010
final data
National Snow and Ice Data Center, University of Colorado Boulder
Total extent = 4.7 million sq km

NSIDC

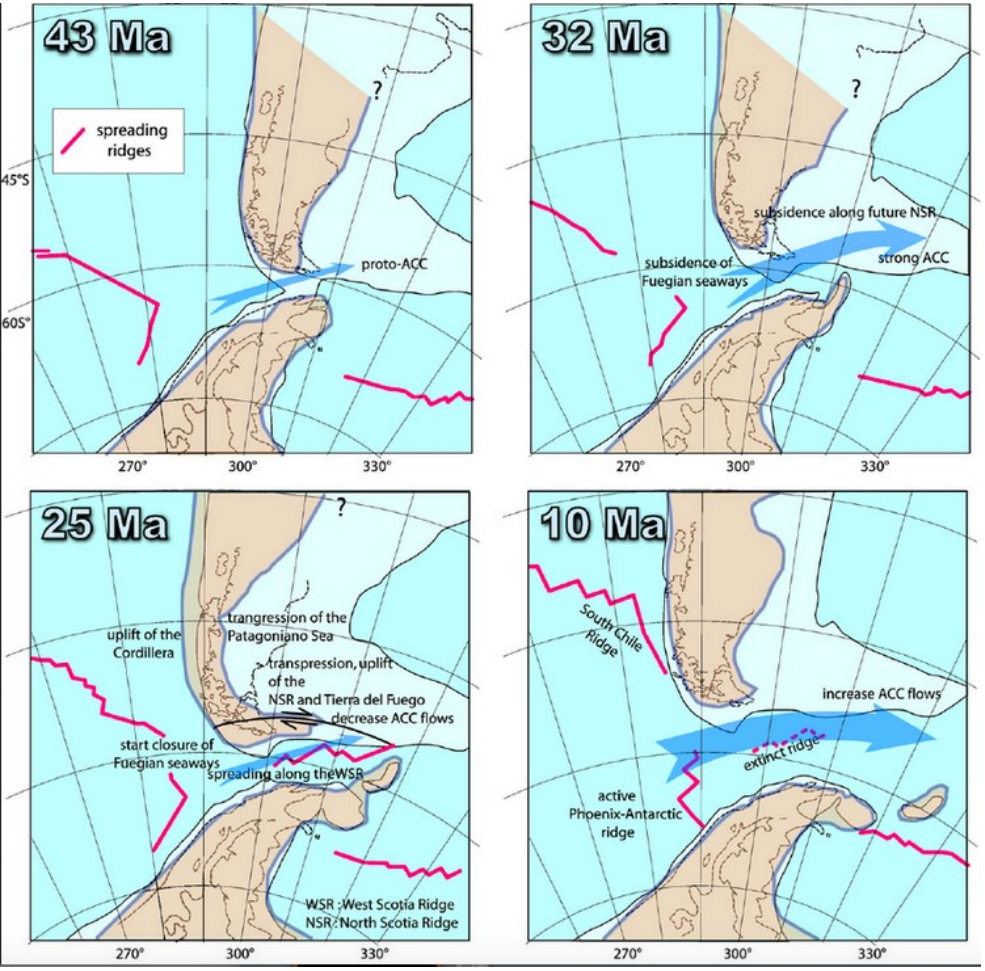
Surface Ocean Current



Opening of Drake Passage
Birth of Antarctica



Opening of Drake Passage
Birth of Antarctica



- Opening of Drake Passage
- Lower meridional ocean heat flux (N→S)
- Increased ocean productivity (CO_2 drawdown)
- Onset of Glaciation

Closing of Isthmus of Panama
Birth of Greenland?





Closing of the Isthmus of Panama

Onset of Gulf Stream

Supply of Moisture for Greenland precipitation

Onset of Greenland glaciation

Photosynthesis Simplified Reaction

Photosynthesis can be described by the simplified chemical reaction

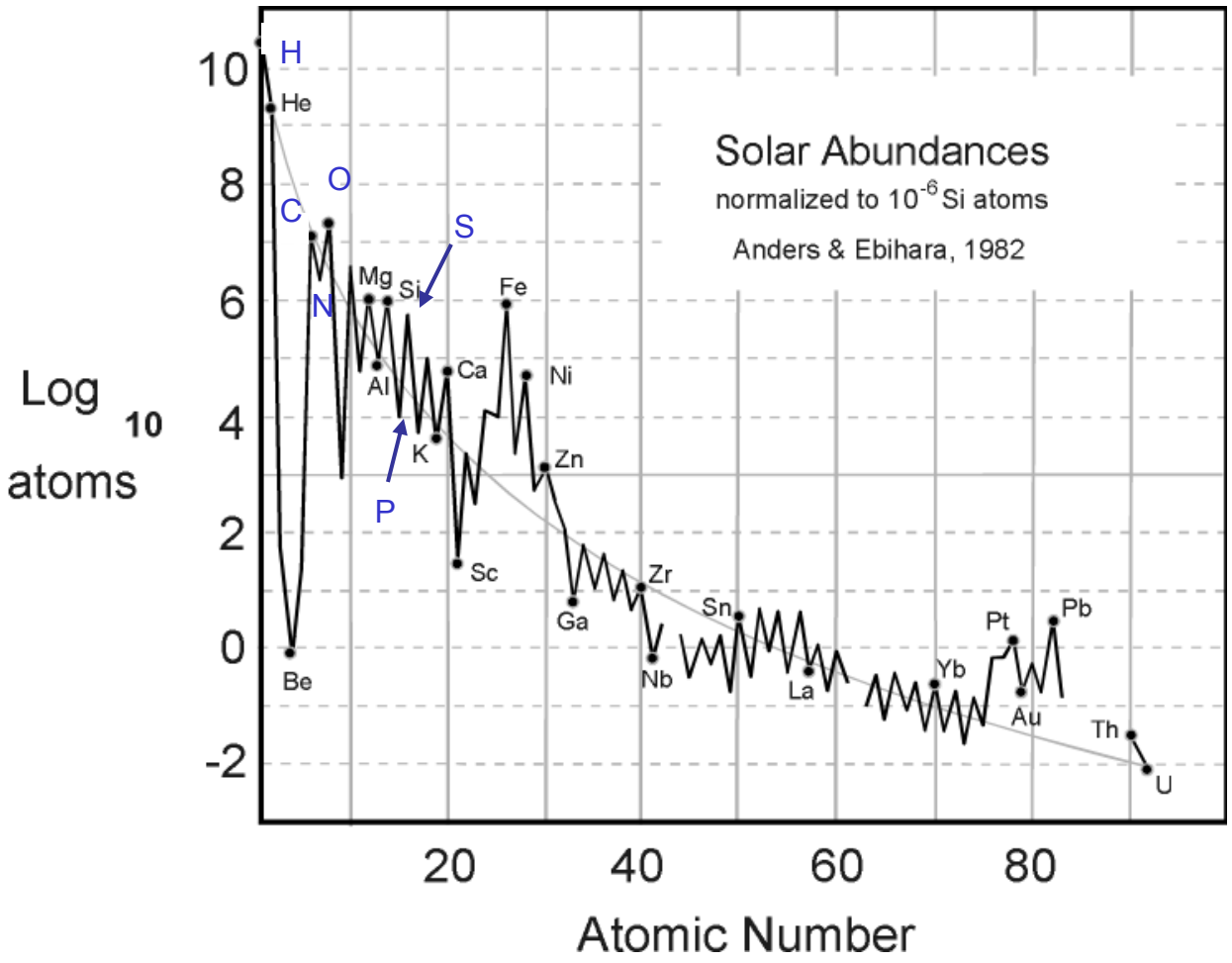
$$6H_2O + 6CO_2 + \text{energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

where $C_6H_{12}O_6$ is glucose (sugar, carbohydrates)

Shortwave radiation
(visible light)



Elements essential for life



Six essential elements:

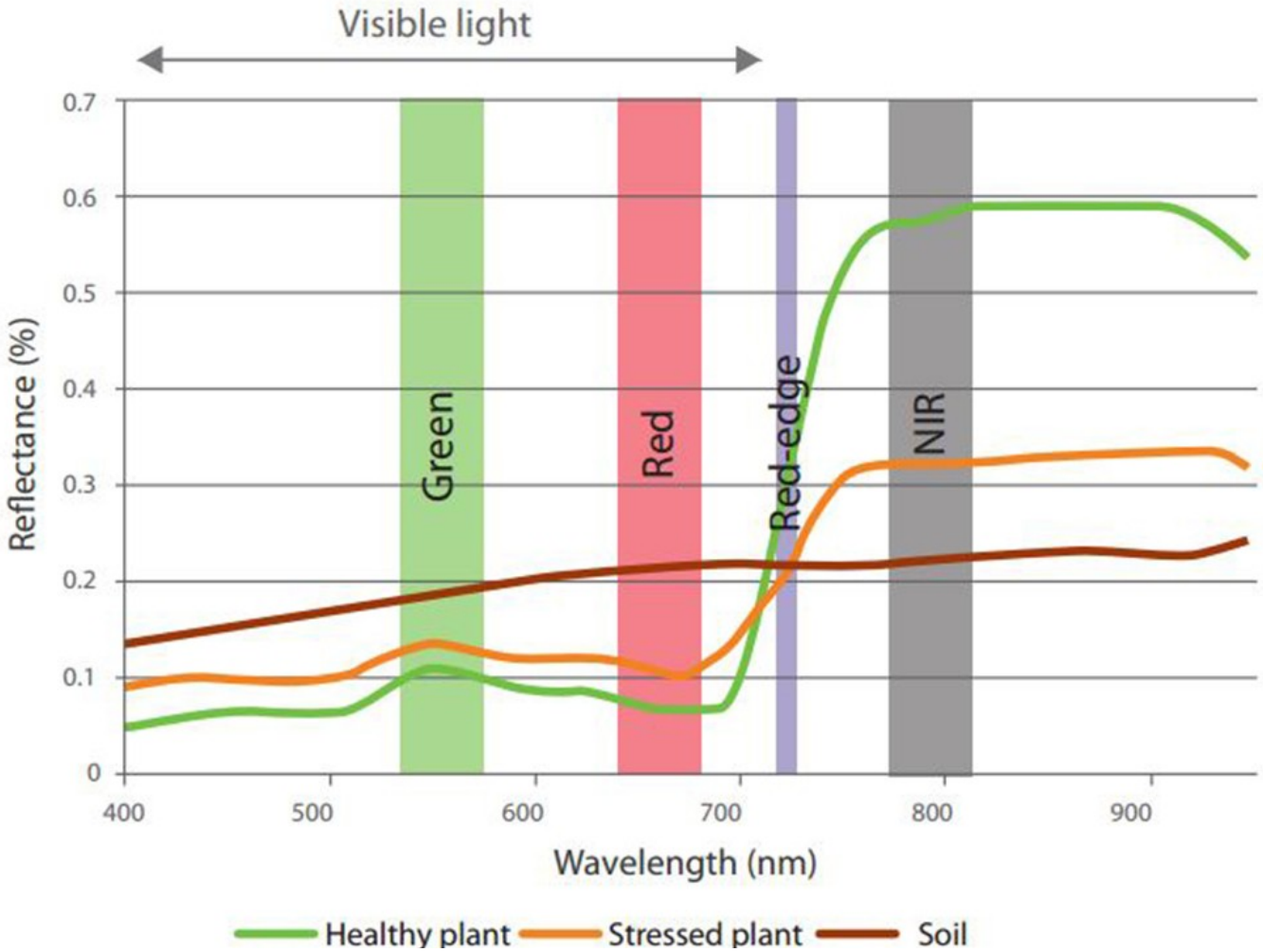
Hydrogen, carbon, nitrogen, oxygen, sulfur, phosphorus

Productivity

The seasonal and spatial patterns of biological activity in the oceans and on the land surface

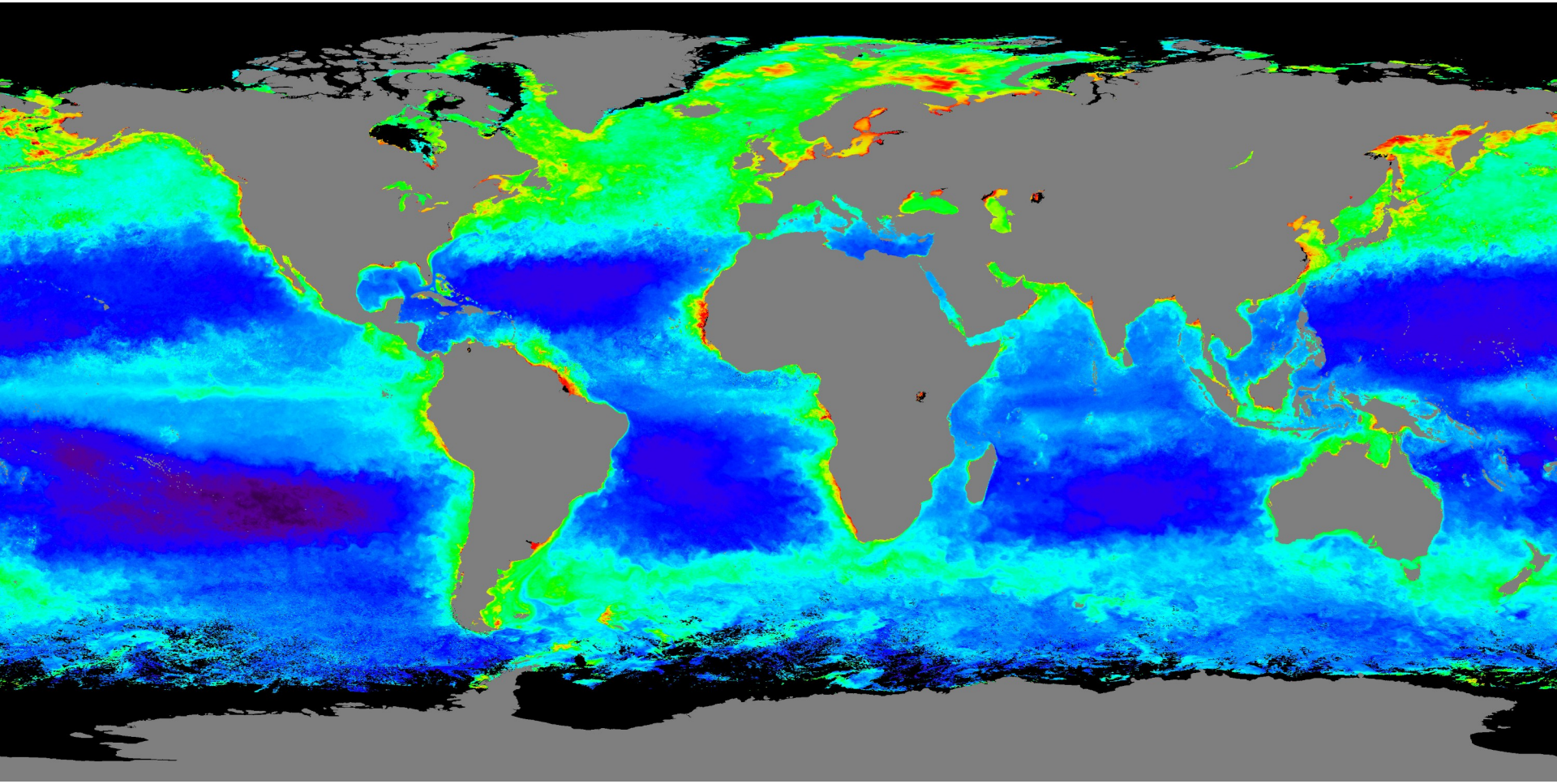
measured by photosynthetic plant or alga biomass production rates (kg/year/area)

Red Edge



Chlorophyll has a unique spectral reflectance

Chlorophyll

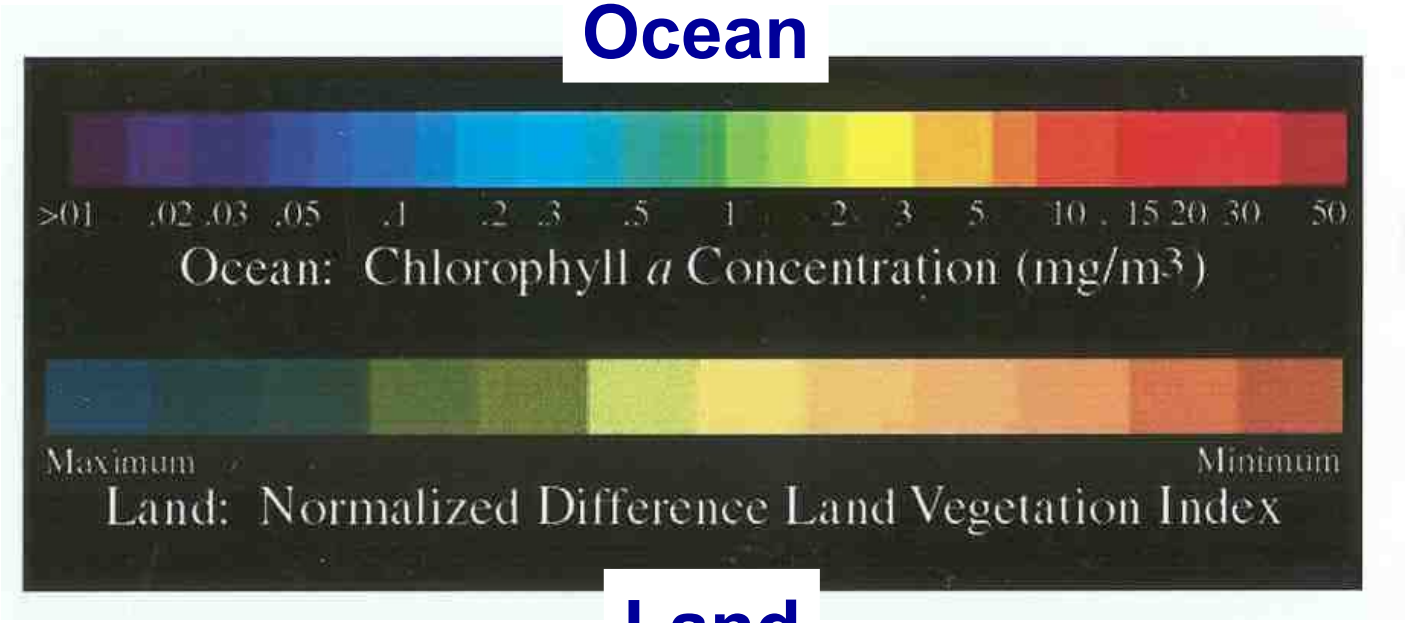


NOAA

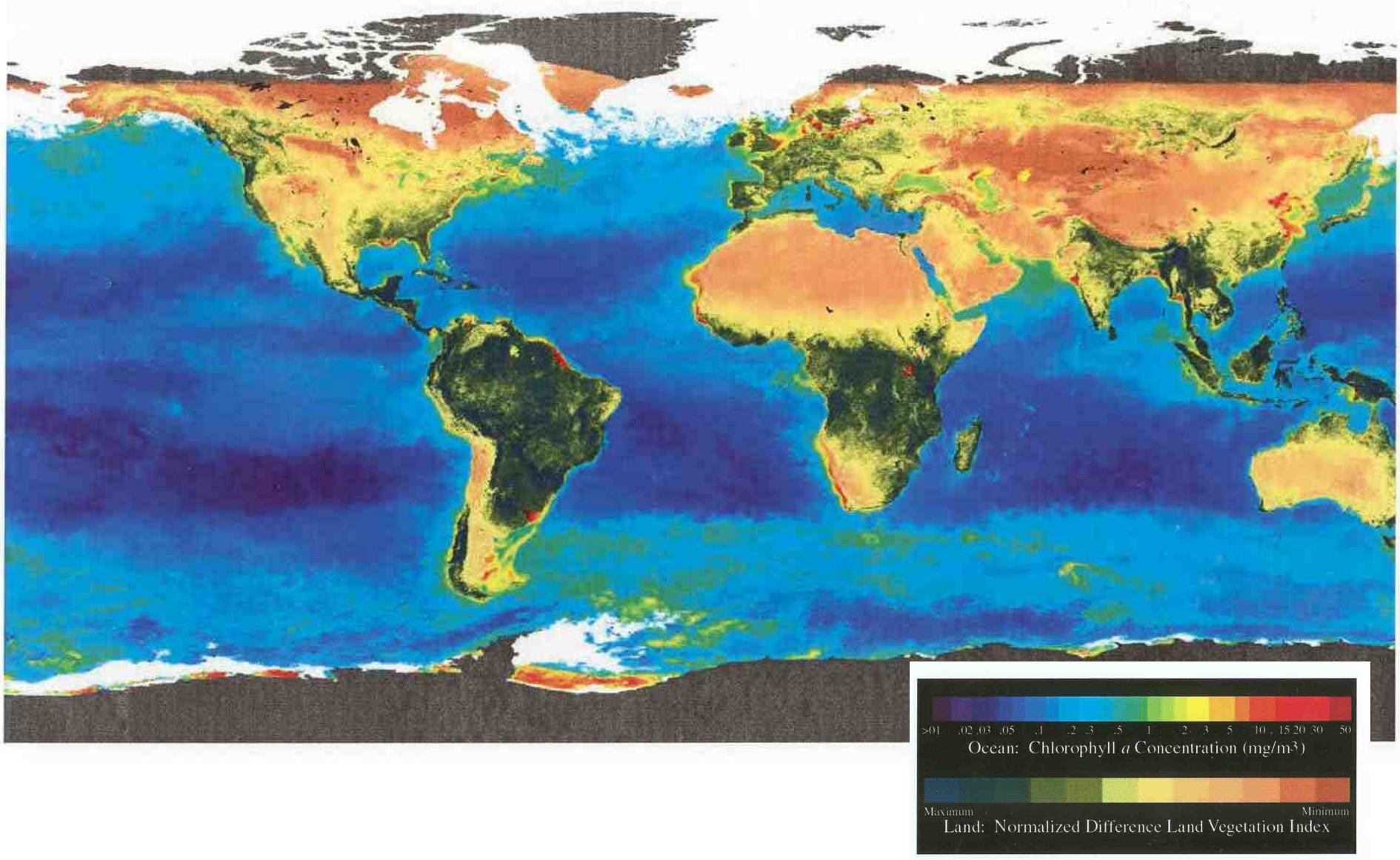
Satellite Data on Productivity

CZCS – Coastal Zone Color Scanner for oceans; detects chlorophyll concentration in water; integrated over ~ 8 yrs

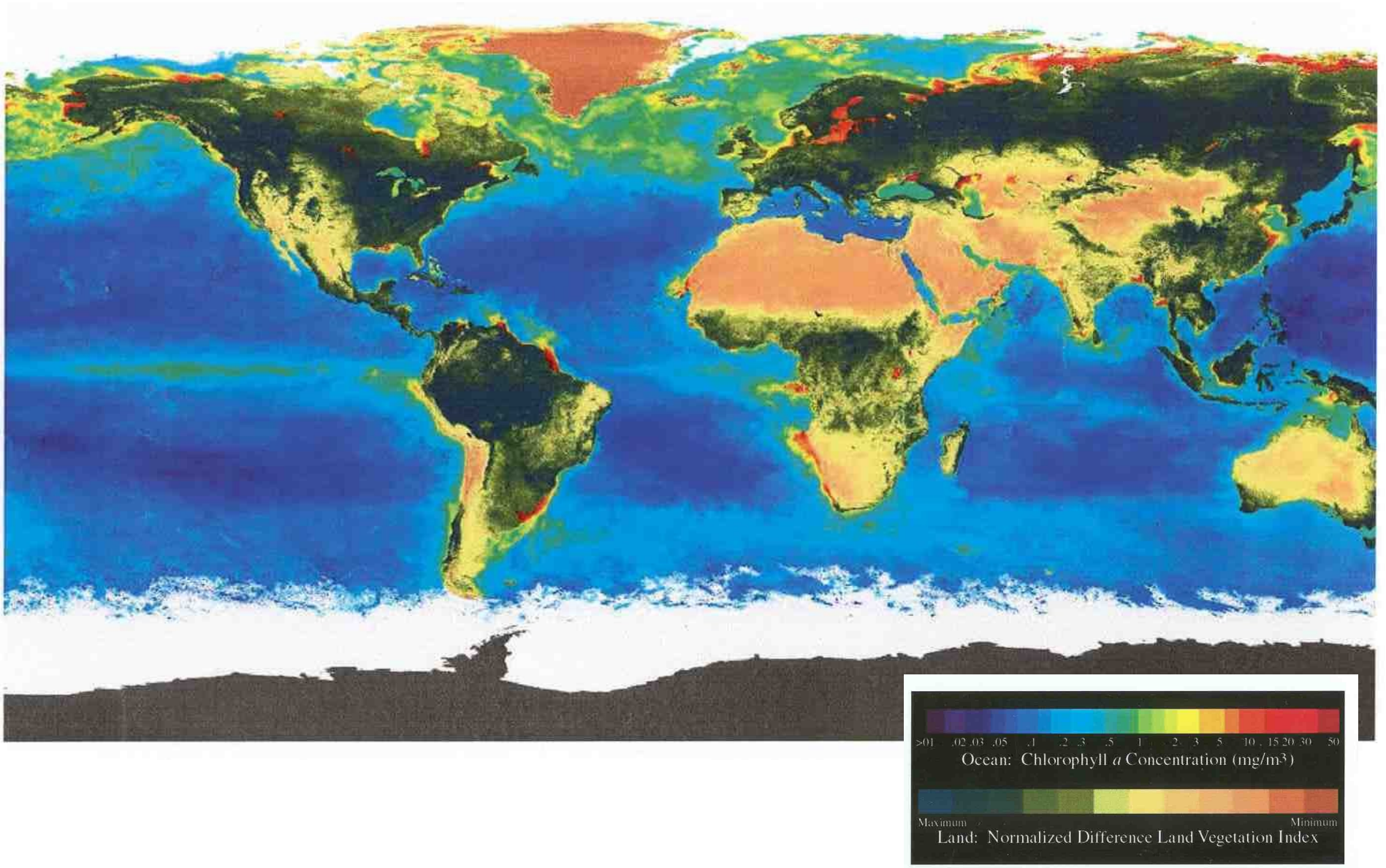
AVHRR – Advanced Very High Resolution Radiometer over land; data collected in visible, near infrared, and thermal infrared regions of spectrum; integrated over ~ 3 yrs



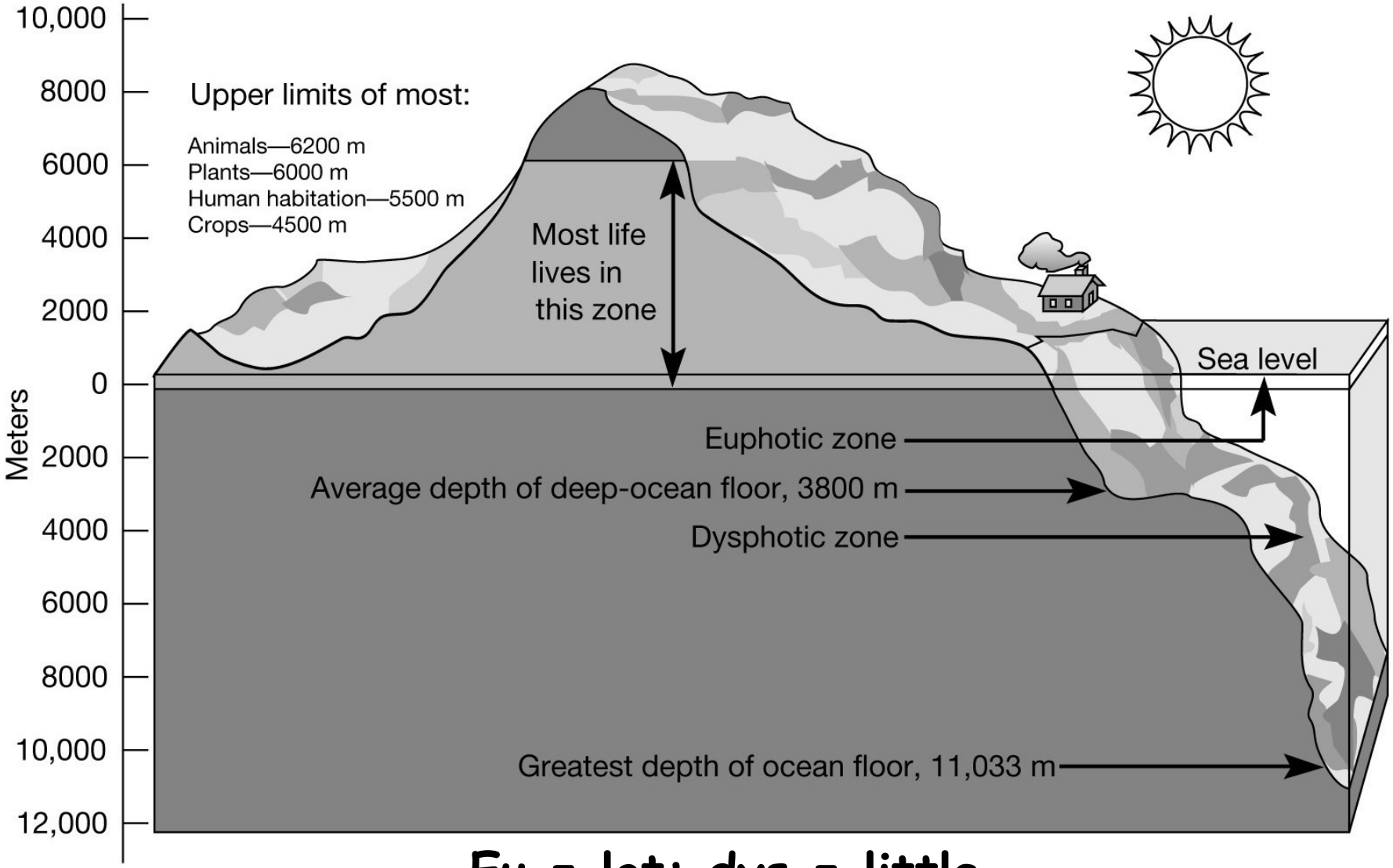
Chlorophyll Distribution and Vegetation Index northern Hemisphere Winter (January)



Chlorophyll Distribution and Vegetation Index northern Hemisphere Summer (July)



Habitable Areas on Earth



Eu = lot; dys = little



dwarf trees near treeline
(in Europe, called ‘Krunholtz’)

Alpine
treeline



Everest from Tibet: 5000 m

Close to limit of human habitation



Vegetation restricted to lichens, shrubs, mosses – adapted to cold, dry, low atmospheric pressure



Close to limit of crop cultivation: barley (green) and rapeseed (yellow) at 4500 m

